V-023222 / L23222 Buduunkhad Exploration Workflow Methodology v5 Explicit Input File Mapped

78 raw input files + QGIS + SNAP 13.0.0 + ASTER workflow + KOMPSAT-2 + ALOS-PALSAR + DJI Matrice 400 + Zenmuse L2/L3/P1 + Olympus Vanta M + Bruker Titan S1 + pXRF + target ranking workflow

Талбар	Утга
Project area	Buduunkhad / XV-023222 / L23222
Standard CRS	WGS 84 / UTM Zone 47N, EPSG:32647
Document type	Methodology / workflow guide. Бодит боловсруулалтын үр дүн, нөөцийн тооцоо, final target баталгаа биш.
Source basis	XV-023222_Buduunkhad_Exploration_Workflow_Methodology_78Inputs_v1.docx дахь 78 raw input file register, CRS, project identity, equipment/software logic.
Style reference	BuduunKhad_Namalzakh_29RawInputs_SNAP_ASTER_Drone_XRF_Methodology.docx-ийн phase format, folder tree, QA/QC, input/output template-ийн бичлэгийн хэв маяг. 29 input file list-ийг хуулж оруулаагүй.
Main principle	Raw data-г өөрчлөхгүй хадгалж, зөвхөн processing copy дээр ажиллана; sidecar metadata файлуудыг parent raster/image-тэй хамт хадгална.
v3 нэмэлт	Historical Scanned Maps QGIS Vectorization Workflow v02 Detailed аргачлалыг Phase 1-ийн дэд аргачлал / Appendix E хэлбэрээр нэмэв.
v4 нэмэлт	Preliminary Deposit Model Preparation Methodology-г Appendix биш, үндсэн workflow-ийн 03_Phase_3 дотор 03A дэд workflow болгон нэгтгэв; Phase 4 preliminary prospect ranking болон Phase 10 final target ranking руу handover холбоос нэмэв.
v5 нэмэлт	Phase бүрийн Input files хэсгийг ерөнхий evidence group-ээр биш, яг ашиглах raw input file № болон filename-аар заасан. Мөн 1A Explicit Input File Assignment Matrix нэмэж, raw input -> primary phase -> methodology action холбоосыг бүрэн тодорхойлов.

Анхааруулга: Satellite, ASTER, KOMPSAT-2, DEM, Drone/LiDAR болон pXRF output нь хүдэржилтийн баталгаа биш. Эдгээр нь target generation, field validation, sampling prioritization-д ашиглах support evidence юм. Эцсийн confidence нь хээрийн шалгалт, лабораторийн шинжилгээ, structural/geological evidence, шаардлагатай бол trench/geophysics/scout drilling-аар баталгаажна.

Агуулгын тойм

0. Methodology scope and governing principles

- Энэхүү v2 Enhanced аргачлал нь Будүүн хад / XV-023222 / L23222 хайгуулын талбайн 78 raw input файлыг professional exploration workflow болгон үе шаттай хэрэгжүүлэх заавар юм.
- Энэ баримт бичиг нь бодит processing result, нөөц/баялгийн тооцоо, эсвэл эцсийн өрөмдлөгийн батлсан target биш. Зөвхөн өгөгдөл боловсруулах, шалгах, талбайд баталгаажуулах, дараагийн шийдвэр гаргах аргачлал юм.
- 78 input file register-ийг хадгалж, файл бүрийг spatial status, limitation, processing action, expected output-тэй холбож шинэчилсэн.
- 29RawInputs аргачлалын input list-ийг ашиглаагүй; зөвхөн формат, QA/QC-ийн бичлэгийн түвшин, folder/output template-ийн бүтэц авсан.

Core principle	Implementation rule
Raw preservation	00_Raw_Files_Archive дотор original file read-only хадгална. Rename хийх шаардлагатай бол standardized name register-д бичиж, raw file-г overwrite хийхгүй.
Processing copy	Бүх боловсруулалтыг 01-11 phase-ийн Input/Working/Processing хавтасны copy дээр хийнэ.
CRS control	Final deliverables EPSG:32647; native CRS/source CRS-г metadata-д хадгална.
Evidence hierarchy	Remote sensing/pXRF/drone = support evidence; lab assay + field geology + structural control = decision evidence.
Decision gates	Phase бүрийн төгсгөлд QA/QC болон go/no-go шалгуураар дараагийн шат руу шилжинэ.

1. Enhanced 78 raw input file register

Доорх хүснэгт нь v1 баримт бичигт байгаа 78 input file-ийн жагсаалтыг хадгалж, хэрэгжүүлэхэд шаардлагатай spatial status, workflow phase, limitation, required processing action, expected output багануудыг нэмсэн register юм. Sidecar файлууд (.tfw, .aux.xml, .ovr, .rpc, .eph, .txt metadata) нь parent raster/image-ээс салгахгүй хадгалагдана.

№	Evidence group	Raw input filename	File type	Spatial status / CRS status	Exploration use	Workflow phase	Confidence / limitation	Required processing action	Expected output
1	01_Tectonic_Terrane_KMZ	Geological and Tectonic Characteristics of the Lake Terrane, Mongolia.png	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Англи эх текст: Lake Terrane-ийн геологи, тектоникийн шинж	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
2	01_Tectonic_Terrane_KMZ	Mongolia_Tectonic_Terrane_Map_Project_Area_Lake_Island_Arc_Terrane.jpg	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Монголын террейний зураг: төслийн талбай Lake island arc terrane-тэй давхцах байдал	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
3	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Central_Mongolian_Massif_and_Daagandel_Tectonic_Zone_Page11.jpg	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	МУГЗ-500 тайлбар, хуудас 11: Дааган дэлийн бүс ба Төв Монголын массив	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
4	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part1_Page08.jpg	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	МУГЗ-500 тайлбар, хуудас 8: Бодонч-Цээлийн бүс, Нуурын Атриат мегабүс	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
5	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part2_Page09.jpg	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	МУГЗ-500 тайлбар, хуудас 9: Хурайн, Баатар хайрханы, Улааншандын бүс	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
6	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part3_Page10.jpg	Зураг/скан	Scanned/non-native image;	МУГЗ-500 тайлба	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until	Scan QA/QC, georeferen	Master inventory and

				georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	р, хуудас 10: Ханхөхийн ба Хантай ширийн бүс		georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	ce if map, digitize key evidence, attribute coding, confidence flag.	confidence ranking entry
7	01_Tectonic_Terrane_KMZ	Regional_Tectonic_Subdivision_Map_of_Mongolia_Tumurtogoo_2017_Buduunkhad_Project_in_Ulaanshand_Zone.jpg	Зураг/скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Монголын тектоник дүүрэгчлэлийн зураг: Бүдүүн хад төслийн талбай Уланшандын бүсэд	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Master inventory and confidence ranking entry
8	01_Tectonic_Terrane_KMZ	MN_BuduunKhad_L23222_LicenseBoundary_WGS84_v01_raw.kmz	KMZ / KML polygon	Spatial polygon/vector; WGS84 байх магадлалтай. QGIS дээр EPSG:4326 -> EPSG:32647 хөрвүүлж баталгаажуулна.	L23222 / Buduunhhad тусгай зөвшөөрлийн хил, WGS84 координатын олон өнцөгт	01_Phase_1_Data_Audit_and_Master_GIS_Setup	Medium/High after CRS, metadata, coordinate and content QA/QC.	Extract coordinates/attributes, clean register, link with GIS layers.	license_boundary layer in Master_GIS_Database.gpkg
9	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_DEM_1arcsec_WGS84_v01_raw.tif	Үндсэн raster өгөгдөл	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	Үндсэн raster өгөгдөл	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
10	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_NumObservations_1arcsec_WGS84_v01_raw.tif	Үндсэн raster өгөгдөл	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	Үндсэн raster өгөгдөл	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
11	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tfw	World file: pixel хэмжээ, байрлал, north-up georeference	Raster sidecar world file; parent raster-тай хамт хадгална, дангаар spatial layer биш.	World file: pixel хэмжээ, байрлал, north-up georeference	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers

12	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif	Үндсэн raster өгөгдөл	GeoTIFF/raster; UTM47N/EPSG:32647 гэж нэрэнд дурдсан боловч CRS, pixel size, extent, NoData-г шалгана.	Үндсэн raster өгөгдөл	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
13	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.aux.xml	Auxiliary metadata: histogram/statistics ба GIS sidecar мэдээлэл	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display support. Parent file-тай хамт хадгална.	Auxiliary metadata: histogram/statistics ба GIS sidecar мэдээлэл	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
14	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.ovr	Overview/pyramid: хурдан харуулахад ашиглагдах багасгасан raster	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display support. Parent file-тай хамт хадгална.	Overview/pyramid: хурдан харуулахад ашиглагдах багасгасан raster	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
15	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tfw	World file: pixel хэмжээ, байрлал, north-up georeference	Raster sidecar world file; parent raster-тай хамт хадгална, дангаар spatial layer биш.	World file: pixel хэмжээ, байрлал, north-up georeference	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
16	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif	Үндсэн raster өгөгдөл	GeoTIFF/raster; UTM47N/EPSG:32647 гэж нэрэнд дурдсан боловч CRS, pixel size, extent, NoData-г шалгана.	Үндсэн raster өгөгдөл	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
17	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.aux.xml	Auxiliary metadata: histogram/statistics ба	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display	Auxiliary metadata: histogram/statistics ба GIS sidecar	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/

			GIS sidecar мэдээлэл	support. Parent file-тай хамт хадгална.	мэдээлэл		Parent raster/image-тэй хамт хадгалах бол High support value.		safety layers
18	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.ovr	Overview/pyramid: хурдан харуулахад ашиглагдах багасгасан raster	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display support. Parent file-тай хамт хадгална.	Overview/pyramid: хурдан харуулахад ашиглагдах багасгасан raster	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
19	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tfw	World file: pixel хэмжээ, байрлал, north-up georeference	Raster sidecar world file; parent raster-тай хамт хадгална, дангаар spatial layer биш.	World file: pixel хэмжээ, байрлал, north-up georeference	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
20	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif	Үндсэн raster өгөгдөл	GeoTIFF/raster; UTM47N/EPSG:32647 гэж нэрэнд дурдсан боловч CRS, pixel size, extent, NoData-г шалгана.	Үндсэн raster өгөгдөл	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
21	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.aux.xml	Auxiliary metadata: histogram/statistics ба GIS sidecar мэдээлэл	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display support. Parent file-тай хамт хадгална.	Auxiliary metadata: histogram/statistics ба GIS sidecar мэдээлэл	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers
22	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.ovr	Overview/pyramid: хурдан харуулахад ашиглагдах	GIS sidecar/overview metadata; parent raster-ийн statistics, pyramid, display support. Parent file-тай	Overview/pyramid: хурдан харуулахад ашиглагдах багасгасан raster	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-	DEM QC, projection check, hillshade/slope/aspect/drainage/terrain derivative.	Terrain derivatives: hillshade, slope, drainage, access/safety layers

			багасгасан raster	хамт хадгална.			тэй хамт хадгалах бол High support value.		
23	03_KOMPSAT2_MSC_L1G	KOMPSAT EULA Form_3.1.pdf	Лицензийн нөхцөл / End User License Agreement	Text/table scanned or office document; coordinate extraction, table cleaning, source confidence log шаардлагатай.	Data provenance, license compliance, тайлангийн хавсралт, хөрөнгө оруулагч/зөвлөх/төрийн байгууллагад эх үүсвэр тайлбарлахад хадгална.	02_Phase_2_Remote_Sensing_Preprocessing	Needs verification.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
24	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.tif	Panchromatic буюу PAN band raster зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Суурь зураг, lineament interpretation, зам/шурф/ухаш/эвдрэл/илгарш ялгах, field route planning, pan-sharpening хийхэд хамгийн чухал үндсэн raster файл.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
25	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.txt	PAN metadata text file	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Тайлангийн Data Source хэсэг, GIS projection/GSD шалгалт, raw data register, QA/QC verification-д ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
26	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.rpc	PAN RPC файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Orthorectification, DEM ашигласан terrain correction, PAN 1 m зургийг газрын бодит байрлалд нарийвчлалтай	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support

					тохируулахад ашиглана.		support value.		
27	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.eph	PAN ephemeris/orbit файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Геометр боловсруулалтын туслах файл; энгийн GIS дээр шууд нээх шаардлагагүй боловч raw bundle-д хадгална.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
28	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.tif	Multispectral Green band raster	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	True color composite-ийн Green component, Red/Blue/NIR band-уудтай хамт RGB/false color composite, гадаргын өнгөний ялгаа, ус/ургамал/илгаршийн ялгаралд ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
29	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.txt	Green band metadata	Spatial/metadata status тодруулах шаардлагатай; Data Audit phase-д шалгана.	Band stacking хийхэд band order шалгах, тайлангийн metadata, QA/QC register-д ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
30	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.rpc	Green band RPC файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Green band orthorectification, бусад MS band болон PAN band-тай spatial alignment хийхэд ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support

31	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.eph	Green band ephemeris/orbit файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	RPC-based геометр засвар, processing audit-д хэрэгтэй; tif/txt/rpc/eph дөрвийг хамтад хадгална.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
32	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.tif	Multispectral Blue band raster	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	True color composite-ийн Blue component, ус/сүүдэр/atmospheric haze/цайвар гадаргын ялгаа, ил гарш ба хөрсний өнгөний ялгааг бусад band-тай харьцуулан тайлна.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
33	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.txt	Blue band metadata	Spatial/metadata status тодруулах шаардлагатай; Data Audit phase-д шалгана.	Blue band source verification, band stacking, тайлангийн хавсралт, data register-д ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
34	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.rpc	Blue band RPC файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Blue band orthorectification, DEM-тэй relief correction, band-to-band alignment хийхэд ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
35	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.eph	Blue band ephemeris/orbit	Sensor/geometric/metadata sidecar; KOMPSAT	Advanced geometric processing	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаа	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт	KOMPSAT orthobasemap / NDVI / lineame

			файл	PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	болон audit trail-д хадгална; шууд зураглал хийх файл биш.		р ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	archive; metadata/RPC audit-д холбох.	nt-outcrop interpretation support
36	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.tif	Multispectral NIR band raster	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	False color composite, vegetation mask, Red band-тай NDVI, ил гарш/ургамлаар хучигдсан хэсэг, drainage pattern, аллювийн ялгаа харахад ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
37	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.txt	NIR band metadata	Spatial/metadata status тодруулах шаардлагатай; Data Audit phase-д шалгана.	NIR band-ийг зөв таних, NDVI/false color/vegetation mask workflow-д source verification хийхэд ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
38	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.rpc	NIR band RPC файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	NIR band orthorectification, бусад band-уудтай давхцуулах, pan-sharpening өмнөх geometry consistency шалгахад хэрэглэнэ.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
39	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.eph	NIR band ephemeris/orbit файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай	Advanced geometric processing, audit trail-д хадгална; шууд interpre	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support

				хамт хадгална.	tation хийх файл биш.		тэй хамт хадгалах бол High support value.		
40	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.tif	Multispectral Red band raster	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	True color composite-ийн Red component, NIR-тэй NDVI, хөрс/илгарш/ferric эсвэл iron-stained гадаргын өнгөний ялгааг ажиглахад ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
41	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.txt	Red band metadata	Spatial/metadata status тодруулах шаардлагатай; Data Audit phase-д шалгана.	Red band source verification, NDVI/true color/false color composite хийхэд band identity баталгаажуулах, QA/QC register-д ашиглана.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
42	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.rpc	Red band RPC файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Red band orthorectification, NIR/Green/Blue/PAN band-уудтай spatial alignment, DEM ашигласан terrain correction-д хэрэгтэй.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
43	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.eph	Red band ephemeris/orbit файл	Sensor/geometric/metadata sidecar; KOMPSAT PAN/MS orthorectification ба audit-д parent raster-тай хамт хадгална.	Геометр боловсруулалтын туслах өгөгдөл; raw data бүртгэлд хадгална.	02_Phase_2_Remote_Sensing_Preprocessing	Support file; parent data-гүй дангаар ашиглахгүй. Parent raster/image-тэй хамт хадгалах бол High support value.	KOMPSAT bundle-ийн parent PAN/MS file-тэй хамт archive; metadata/RPC audit-д холбох.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support

44	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jpg	Browse image / preview зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Scene coverage зөв эсэхийг хурдан шалгах, data register/catalog thumbnail, тайлангийн internal preview-д ашиглана. Analysis хийх үндсэн зураг биш.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
45	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jgw	Browse image world file	Raster sidecar world file; parent raster-тай хамт хадгална, дангаар spatial layer биш.	br.jpg-г ArcGIS/QGIS дээр ойролцоогоор зөв байрлалтай нээх, lightweight preview overlay хийхэд ашиглана. Дангаараа зураг биш.	02_Phase_2_Remote_Sensing_Preprocessing	Needs verification.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
46	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_tn.jpg	Thumbnail image	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Data inventory, thumbnail preview, тайлангийн хавсралтад ашиглаж болно. Geospatial analysis, pixel value analysis хийхэд ашиглахгүй.	02_Phase_2_Remote_Sensing_Preprocessing	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	PAN/MS import, band identity check, RPC alignment, orthorectification, pan-sharpen/visual products.	KOMPSAT orthobasemap / NDVI / lineament-outcrop interpretation support
47	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_1-200000_v01_raw-scan.jpg	Зураг / шлихийн сорьцлолтын үр дүн	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Шлихийн сорьц, ёроолын сорьц, ашигт малтмалын/индикатор минералын тархалтын контур, элемент-минералын тэмдэглэгээ.	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Georeference/digitize anomaly contours, drainage catchment analysis, field follow-up planning.	stream_sediment_anomaly and heavy_mineral_anomaly GIS layers; follow-up plan
48	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	Таних тэмдэг /	Scanned/non-native image; georefere	Шлих, ёроолын сорьцы	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Medium/Low until georef	Georeference/digitize anomaly contours,	stream_sediment_anomaly

			шлихийн зураг	nce status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	н тэмдэг, минералуудын нэршил, агуулгын ангилал, контурын тайлбар.		erenced and source checked; scale/residual/confidence flag шаардлагатай.	drainage catchment analysis, field follow-up planning.	and heavy_mineral_anomaly GIS layers; follow-up plan
49	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	Таних тэмдэг / ёоролын сорьц	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Ёоролын сорьцын агуулга, сарнилын урсгал, сав газрын контур, дээжлэлт хийсэн судлаачдын жагсаалт.	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Georeference/digitize anomaly contours, drainage catchment analysis, field follow-up planning.	stream_sediment_anomaly and heavy_mineral_anomaly GIS layers; follow-up plan
50	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Polyelement_1-200000_v01_raw-scan.jpg	Зураг / ёоролын сорьцын полиметалл үр дүн	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Cu, Pb, Zn, Ag, As, Bi, W, Sn, Mo, Mn, Ba, F зэрэг олон элементийн сарнилын хүрээ ба урсгалын зураглал.	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Georeference/digitize anomaly contours, drainage catchment analysis, field follow-up planning.	stream_sediment_anomaly and heavy_mineral_anomaly GIS layers; follow-up plan
51	04_HeavyMineral_StreamSediment_Field	2011_MN_Namalzakh_L47-74-A_FieldRouteNotebook_Routes14-15_Obs1076-1090_1-50000_v01_raw-scan.pdf	PDF / хээрийн маршрутын дэвтэр	Text/table scanned or office document; coordinate extraction, table cleaning, source confidence log шаардлагатай.	Маршрут 14, 15; ажиглалтын цэг 1076-1090; гар зураг, координат, чулуулгийн гарш, судал, структурын тэмдэглэл.	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Georeference/digitize anomaly contours, drainage catchment analysis, field follow-up planning.	stream_sediment_anomaly and heavy_mineral_anomaly GIS layers; follow-up plan
52	04_HeavyMineral_StreamSediment_Field	XV023222_Buduunkhad_Report7255_FieldObservation_StationDescription_Table_WGS84_LegacyRaw_v01.pdf	PDF / ажиглалтын цэгийн хүснэгт	Text/table scanned or office document; coordinate extraction, table cleaning, source confidence log шаардлагатай.	Ажиглалтын цэгүүдийн координат, чулуулаг, структур, эрдэсжилтийн товч тайлбар.	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	Needs verification.	Georeference/digitize anomaly contours, drainage catchment analysis, field follow-up planning.	stream_sediment_anomaly and heavy_mineral_anomaly GIS layers; follow-up plan
53	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_1-200000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown.	Региональ геологийн суурь зураг	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and	Scan QA/QC, georeference if map, digitize key evidence,	Geology/mineral occurrence/prospectivity layers

				Map grid/GCP ашиглан QA/QC хийх шаардлагатай.			source checked; scale/residual/confidence flag шаардлагатай.	attribute coding, confidence flag.	and occurrence database
54	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_Legend_1-200000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Региональ геологийн зурагт ашигласан стратиграфи, интрузив, структур, литологийн тайлбар	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
55	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Нарийвчилсан геологи, литологи, хагарал, зүсэлт	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
56	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_Legend_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Нарийвчилсан стратиграфи, интрузив, судал, хувирлын тэмдэглэгээ	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
57	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_1-200000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Региональ ашигт малтмалын илрэл, геохимийн гажил, хүдрийн талбай	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
58	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_Legend_1-200000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Ашигт малтмал, элемент, илрэл, гажлын тэмдэглэгээ	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database

							flag шаардлагатай.		
59	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralDistributionPatternMap_1-100000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Металлогенийн бүс, хүдрийн дүүрэг, зангилаа, талбайн харилцан байрлал	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
60	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralOccurrenceMap_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Au-Cu, Cu, Mo, As, Zn зэрэг илрэл ба геологийн суурьтай давхцуулсан зураг	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
61	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MetallogenicSchemeAndMetallogenogram_1-400000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Хүдрийн формац, нас, металлогенийн бүс ба элементүүдийн холбоо	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
62	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessment_ReportExcerpt_B3-TolKhar_v01_raw-photo.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Б-2, Б-3, Б-4 талбайн тайлбар, талбайн хэмжээ, илрэл, хэтийн төлөв	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
63	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessmentMap_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Б-3 Толь хяр, Г-1 зэрэг хэтийн төлөвтэй хэсгийг ялгасан зураг	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
64	05_Geology_Mineral_	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image;	Маршрут, ажигла	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until	Scan QA/QC, georeferen	Geology/mineral occurre

	Prospectivity			georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	лтын цэг, сорьц, суваг, шурф, зүсэлтийн байрлал		georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	ce if map, digitize key evidence, attribute coding, confidence flag.	nce/prospectivity layers and occurrence database
65	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_Legend_1-50000_v01_raw-scan.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Ажиглалтын цэг, маршрут, сорьц, шурф, суваг, тэмдэглэгээ	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
66	05_Geology_Mineral_Prospectivity	2013_MN_L47-XIX_GoldOccurrenceDescriptions_XIX-74-A_4186_v01_raw.docx	DOCX текст	Text/table scanned or office document; coordinate extraction, table cleaning, source confidence log шаардлагатай.	XIX-74-A-1, A-2, A-3 алтны илрэлийн координат, агуулга, геологийн нөхцөл	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/High after CRS, metadata, coordinate and content QA/QC.	Extract coordinates/attributes, clean register, link with GIS layers.	Geology/mineral occurrence/prospectivity layers and occurrence database
67	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralOccurrenceAndMineralizedPointRegister_7255_v01_raw-scan.pdf	PDF скан, 14 хуудас	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Илрэл, эрдэсжсэн цэг, гар зураг, хүснэгт, РЗ тайлбар	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Geology/mineral occurrence/prospectivity layers and occurrence database
68	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralizedPointRegister_7255_v01_raw-table.xlsx	XLSX хүснэгт	Text/table scanned or office document; coordinate extraction, table cleaning, source confidence log шаардлагатай.	Цэгийн дугаар, координат, чулуулаг, агуулга, эрдэсжилт, төрөл, дээж	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/High after CRS, metadata, coordinate and content QA/QC.	Extract coordinates/attributes, clean register, link with GIS layers.	Geology/mineral occurrence/prospectivity layers and occurrence database
69	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_Legend_RawScan_2020_v01.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Металлогений зургийн таних тэмдэг, ашигт малтмалын төрөл, орд-илрэлийн тэмдэглэгээ, судалга	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Metallogenic context map/report and deposit model screening table

					аны ажлын тэмдэглэгээ		шаардлагатай.		
70	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_RawScan_2020_v01.jpg	JPG зураг	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Монгол улсын 1:500,000-ны металлогений зураг, L-47-B (Тал шанд) хавтгай	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Metallogenic context map/report and deposit model screening table
71	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book01_ProjectBook13_1M500K_RawScan_2021_v01.pdf	PDF скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Монгол орны 1:500,000-ны масштабын металлогений зураг, тайлбар бичиг. Ном-1, төслийн номын дугаар-13	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Metallogenic context map/report and deposit model screening table
72	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book04_ProjectBook16_1M500K_RawScan_2021_v01.pdf	PDF скан	Scanned/non-native image; georeference status unknown. Map grid/GCP ашиглан QA/QC хийх шаардлагатай.	Монгол орны 1:500,000-ны масштабын металлогений зураг, тайлбар бичиг. Ном-4, төслийн номын дугаар-16	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis	Medium/Low until georeferenced and source checked; scale/residual/confidence flag шаардлагатай.	Scan QA/QC, georeference if map, digitize key evidence, attribute coding, confidence flag.	Metallogenic context map/report and deposit model screening table
73	07_Basemap_Sentinel2_ASTER	2005-09-05_MN_ASTER-L1B_MultispectralImagery_00409052005043503_v01_raw.hdf	HDF4	ASTER HDF raw product; direct GIS layer биш. ASTER workflow v5 дагуу import/band extraction шаардлагатай.	error	02_Phase_2_Remote_Sensing_Preprocessing	Needs verification; HDF import алдаа/compatibility шалгана. Raw product өөрчлөхгүй.	ASTER HDF import, band extraction, UTM47 grid, index/score/class/mask workflow.	ASTER raw score, class map, final binary support mask, QA/QC log
74	07_Basemap_Sentinel2_ASTER	2025-05-28_MN_T46TGS_GeoreferencedSatelliteRaster_v01_raw.tif	GTiff	GeoTIFF/raster; UTM46N/T46 tile байж болзошгүй. EPSG:32647 стандартруу reproject хийх эсэхийг шалгана.	ok	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	Remote sensing QA/QC, subset/reproject, derive composites/indices, export EPSG:32647.	Sentinel/ASTER/basemap derivative GeoTIFF and remote sensing QA/QC report
75	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_GoogleMaps_BasemapImagery_RGB_2p4m_WGS84_Raw_v01.tif	GTiff	GeoTIFF/raster; CRS/resolution/extent/NoData/band	ok	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordi	Remote sensing QA/QC, subset/reproject, derive composite	Sentinel/ASTER/basemap derivative GeoTIF

				count шалгана.			nate and content QA/QC.	s/indices, export EPSG:32647.	F and remote sensing QA/QC report
76	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_HighResolution_RGB_SurfaceBasemap_GoogleMaps_EPSG3857_0p15m_Raw_v01.tif	GTiff	GeoTIFF/raster; CRS/resolution/extent/NoData/band count шалгана.	ok	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	Remote sensing QA/QC, subset/reproject, derive composites/indices, export EPSG:32647.	Sentinel/ASTER/basemap derivative GeoTIFF and remote sensing QA/QC report
77	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_GeologicalInterpretation_RGB_B12-B08-B03_10m_UTM46N_ReceivedRaw_v01.tif	GTiff	GeoTIFF/raster; UTM46N/T46 tile байж болзошгүй. EPSG:32647 стандарт руу reproject хийх эсэхийг шалгана.	ok	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	Remote sensing QA/QC, subset/reproject, derive composites/indices, export EPSG:32647.	Sentinel/ASTER/basemap derivative GeoTIFF and remote sensing QA/QC report
78	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_LithologyIndex_B11B12_B08B11_B04B03_10m_UTM46N_ReceivedRaw_v01.tif	GTiff	GeoTIFF/raster; UTM46N/T46 tile байж болзошгүй. EPSG:32647 стандарт руу reproject хийх эсэхийг шалгана.	ok	02_Phase_2_Remote_Sensing_Preprocessing	Medium/High after CRS, metadata, coordinate and content QA/QC.	Remote sensing QA/QC, subset/reproject, derive composites/indices, export EPSG:32647.	Sentinel/ASTER/basemap derivative GeoTIFF and remote sensing QA/QC report

Evidence group	File count	Primary workflow use
01_Tectonic_Terrane_KMZ	8	Terrane, tectonic setting, license boundary and regional concept.
02_DEM_ALOS_ASTERGDEM	14	Terrain, slope, drainage, access, safety and lineament support.
03_KOMPSAT2_MSC_L1G	24	High-resolution visual basemap, PAN/MS, lineament/outcrop/access interpretation.
04_HeavyMineral_StreamSediment_Field	6	Historical geochemical anomaly, drainage follow-up and orientation sampling.
05_Geology_Mineral_Prospectivity	16	Detailed/regional geology, occurrences, prospectivity and occurrence database.
06_Regional_Metallogenic_L47B	4	1:500,000 metallogenic context and deposit model screening.
07_Basemap_Sentinel2_ASTER	6	Sentinel/ASTER/basemap remote sensing support layers.

1A. Explicit raw input assignment by workflow phase

Purpose: Энэ хэсэг нь 78 raw input file бүрийг яг аль workflow phase-ийн input болохыг, filename-ээр нь, аргачлалын хэрэглээтэй нь холбож заана. Phase дотор “Input files” гэж ерөнхий бичихгүй; доорх № болон filename-ийг ашиглана. Raw data-г өөрчлөхгүй, зөвхөн working copy дээр боловсруулна.

1A.1 Phase-level input control summary

Workflow phase	Exact direct raw input №	How to apply	Handover note

00 Raw Files Archive	№1-78	Бүх raw file болон sidecar файлыг read-only archive-д бүртгэж checksum үүсгэнэ.	All later phases use working copies only.
01 Data Audit and Master GIS Setup	№1-78	Файл бүрийн metadata, CRS/spatial status, sidecar, scan quality, processing copy status, confidence-г шалгана.	Master inventory + Master GIS schema.
02 Remote Sensing Preprocessing	№9-46, №73-78	DEM, KOMPSAT, ASTER, Sentinel/basemaps-ийг QA/QC, orthorectify/reproject/subset/derive products болгоно.	Support evidence only; ore proof биш.
03 Geological, Metallogenic and CMCS Synthesis	№1-8, №53-72	Tectonic, geology, occurrence, prospectivity, source material, metallogenic context-ийг нэгтгэнэ.	Deposit model and geological evidence layers.
03A Preliminary Deposit Model Preparation	№1-8, №47-78, №9-46 as support	Ордын candidate model бүрт exact source evidence ашиглан supporting/missing/validation хүснэгт гаргана.	Phase 4 scoring and Phase 10 final target model-fit.
04 Preliminary Prospect Delineation and Ranking	Traceable basis №1-78 + Phase 1-3 outputs	Evidence overlay, 100-point score, A/B/C/D ranking; dominant_deposit_model талбар нэмнэ.	A/B prospects to drone/recon.
05 Drone/LiDAR/Photogrammetry	№8, №9-22, №24-46, №75-78 + Phase 4 outputs	Flight block, terrain/access/safety/basemap/lineament support.	Drone outputs to Phase 6-10.
06 Recon Mapping and pXRF	№8, №9-22, №55-56, №60, №63-68, №75-78 + Phase 4/5 outputs	Field validation, lithology/alteration/mineralization/pXRF forms and traverse planning.	Validated field evidence to Phase 7.
07 Rock Chip/Channel Sampling	№52, №55-56, №60, №63-68, №9-22, №75-78 + Phase 6 outputs	Sample candidate selection, lab submission, QA/QC insertion, assay import template.	Lab evidence to Phase 10.
08 Orientation Soil/Stream/Heavy Mineral	№47-52, №9-22, №53-56, №60, №63-64, №68	Historical drainage/heavy mineral/geochemical evidence, orientation survey design.	Validated method to Phase 9.
09 Systematic Soil Grid	№8, №9-22, №47-52, №55, №60, №63-64, №68, №75-78 + Phase 8 outputs	Grid design, soil collection, lab QA/QC, soil anomaly map.	Assay anomalies to Phase 10.
10 Integrated Interpretation and Final Target Ranking	Full traceable source basis №1-78 + validated phase outputs	All evidence, field/lab QA/QC, model-fit re-score, final target sheets.	Final A/B targets to Phase 11.
11 Follow-up Trench/Geophysics/Scout Drill	№8, №9-22, №55, №60, №63-64, №68, №75-78 + Phase 10 outputs	Trench/IP/magnetic/scout drill collar planning, HSE/access/budget check.	Only if minimum criteria met.
99 Final Deliverables	№1-78 traceability + all phase outputs	Final package, GIS, reports, QA/QC, limitations, source references.	Ready for technical/management review.

1A.2 File-by-file input assignment matrix

Энэ matrix-д raw input file бүрийг primary workflow phase-тэй холбов. Secondary phase-үүдэд ашиглах үед тухайн phase-ийн “Input files” мөр болон Phase-level summary-д заасан №-ээр татаж хэрэглэнэ.

№	Evidence group	Exact raw input filename	Primary input phase	Methodology action
1	01_Tectonic_Terrane_KMZ	Geological and Tectonic Characteristics of the Lake Terrane, Mongolia.png	03	Lake Terrane tectonic/geological context; use as regional concept evidence only after source and scan QA/QC.
2	01_Tectonic_Terrane_KMZ	Mongolia_Tectonic_Terrane_Map_Project_Area_Lake_Island_Arc_Terrane.jpg	03	Terrane overlay; supports Lake island arc terrane context and deposit model screening.
3	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Central_Mongolian_Massif_and_Daagandel_Tectonic_Zone_Page11.jpg	03	Regional tectonic explanation; extract narrative evidence and confidence/source note.
4	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part1_Page08.jpg	03	Nuur accretionary megazone context; use for regional geological setting.
5	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part2_Page09.jpg	03	Khurai/Baatarkhairkhan/Ulaanshand zone context; supports model screening.
6	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part3_Page10.jpg	03	Khankhukhi/Khantayshir regional context; source note and limitation flag required.

7	01_Tectonic_Terrane_KMZ	Regional_Tectonic_Subdivision_Map_of_Mongolia_Tumurtogoo_2017_Buduunkhad_Project_in_Ulaanshand_Zone.jpg	03	Ulaanshand Zone tectonic subdivision context for deposit model candidate screening.
8	01_Tectonic_Terrane_KMZ	MN_BuduunKhad_L23222_LicenseBoundary_WGS84_v01_raw.kmz	01	Primary license boundary; import to GeoPackage, reproject EPSG:32647, use for all clipping/buffer/QA.
9	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_DEM_1arcsec_WGS84_v01_raw.tif	02	DEM terrain base; derive hillshade/slope/aspect/drainage and use for access/safety/lineament support.
10	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_NumObservations_1arcsec_WGS84_v01_raw.tif	02	ASTER GDEM observation-count QA layer; evaluate DEM reliability/artefacts.
11	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tfw	02	World file sidecar for ALOS-PALSAR DEM; keep with parent raster, do not use alone.
12	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif	02	High-resolution terrain DEM; produce DTM derivatives, drainage, slope and lineament support.
13	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.aux.xml	02	Auxiliary metadata/statistics sidecar for ALOS DEM; preserve with parent raster.
14	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.ovr	02	Overview/pyramid sidecar for ALOS DEM; preserve with parent raster.
15	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tfw	02	World file sidecar for hillshade; keep with parent raster.
16	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif	02	Hillshade support layer; use for structural/terrain interpretation and field/drone planning.
17	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.aux.xml	02	Auxiliary metadata/statistics sidecar for hillshade; preserve with parent raster.
18	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.ovr	02	Overview/pyramid sidecar for hillshade; preserve with parent raster.
19	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tfw	02	World file sidecar for slope raster; keep with parent raster.
20	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif	02	Slope degree raster; use for access, safety, drainage and drone/trench/drill workability.
21	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.aux.xml	02	Auxiliary metadata/statistics sidecar for slope raster; preserve with parent raster.
22	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.ovr	02	Overview/pyramid sidecar for slope raster; preserve with parent raster.
23	03_KOMPSAT2_MSC_L1G	KOMPSAT EULA Form_3.1.pdf	02	License/provenance evidence; store in data source and compliance register.
24	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.tif	02	KOMPSAT PAN raster; orthorectify/pansharpen, use for lineament, outcrop, access and basemap.
25	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.txt	02	PAN metadata; use for source verification, GSD/projection audit, processing register.
26	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.rpc	02	PAN RPC sidecar; required for orthorectification/terrain correction.

№	Evidence group	Exact raw input filename	Primary input phase	Methodology action
27	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.eph	02	PAN ephemeris/orbit sidecar; preserve with

				PAN bundle for geometric audit.
28	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.tif	02	KOMPSAT Green band; stack with MS bands for true/false color and pan-sharpened products.
29	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.txt	02	Green band metadata; verify band identity and processing parameters.
30	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.rpc	02	Green band RPC sidecar; use for MS orthorectification/alignment.
31	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.eph	02	Green band ephemeris/orbit sidecar; preserve with MS bundle.
32	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.tif	02	KOMPSAT Blue band; stack with MS bands for composites and basemap.
33	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.txt	02	Blue band metadata; verify source and band identity.
34	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.rpc	02	Blue band RPC sidecar; use for MS orthorectification/alignment.
35	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.eph	02	Blue band ephemeris/orbit sidecar; preserve with MS bundle.
36	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.tif	02	KOMPSAT NIR band; derive NDVI/vegetation mask and false color support.
37	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.txt	02	NIR band metadata; verify source and band identity.
38	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.rpc	02	NIR band RPC sidecar; use for MS orthorectification/alignment.
39	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.eph	02	NIR band ephemeris/orbit sidecar; preserve with MS bundle.
40	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.tif	02	KOMPSAT Red band; stack for RGB/NDVI/false color support.
41	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.txt	02	Red band metadata; verify source and band identity.
42	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.rpc	02	Red band RPC sidecar; use for MS orthorectification/alignment.
43	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.eph	02	Red band ephemeris/orbit sidecar; preserve with MS bundle.
44	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jpg	02	KOMPSAT browse image; use for scene coverage preview only, not analysis-grade raster.
45	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jgw	02	Browse image world file; keep with br.jpg for lightweight overlay only.
46	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_tn.jpg	02	Thumbnail preview; use for inventory/catalog only.
47	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_1-200000_v01_raw-scan.jpg	08	Heavy mineral sampling result map; georeference/digitize indicator minerals/anomaly polygons and drainage follow-up.
48	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	08	Heavy mineral legend; build symbol dictionary and indicator mineral domain values.
49	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	08	Stream sediment legend; build element/anomaly/contour domain values.
50	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Polyelement_1-200000_v01_raw-scan.jpg	08	Stream sediment polyelement map; digitize Cu-Pb-Zn-Ag-As-Bi-W-S

				n-Mo-Mn-Ba-F anomaly layers.
51	04_HeavyMineral_StreamSediment_Field	2011_MN_Namalzakh_L47-74-A_FieldRouteNotebook_Routes14-15_Obs1076-1090_1-50000_v01_raw-scan.pdf	08	Field route notebook; extract observation routes/points, compare with geology and sampling plans.
52	04_HeavyMineral_StreamSediment_Field	XV023222_Buduunkhad_Report7255_FieldObservation_StationDescription_Table_WGS84_LegacyRaw_v01.pdf	08	Field observation station table; extract coordinates/lithology/mineralization into validation register.

№	Evidence group	Exact raw input filename	Primary input phase	Methodology action
53	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_1-200000_v01_raw-scan.jpg	03	Regional geology map; georeference and digitize regional geology/structure for context.
54	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_Legend_1-200000_v01_raw-scan.jpg	03	Regional geology legend; build lithology/age/intrusion/structure lookup.
55	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_1-50000_v01_raw-scan.jpg	03	Detailed geology map; primary local geology/structure/contact/section vectorization source.
56	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_Legend_1-50000_v01_raw-scan.jpg	03	Detailed geology legend; build stratigraphy, intrusive, vein and alteration domains.
57	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_1-200000_v01_raw-scan.jpg	03	Regional mineral resources map; digitize occurrence/resource/anomaly context.
58	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_Legend_1-200000_v01_raw-scan.jpg	03	Mineral resources legend; build commodity/occurrence/anomaly symbol dictionary.
59	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralDistributionPatternMap_1-100000_v01_raw-scan.jpg	03	Mineral distribution pattern map; ore district/node/metallogenic area context.
60	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralOccurrenceMap_1-50000_v01_raw-scan.jpg	03	Detailed mineral occurrence map; primary Au-Cu/Cu/Mo/As/Zn occurrence vectorization source.
61	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MetallogenicSchemeAndMetallogenogram_1-400000_v01_raw-scan.jpg	03	Metallogenic scheme/metallogenogram; ore formation and regional metallogenic context.
62	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessment_ReportExcerpt_B3-TolKhar_v01_raw-photo.jpg	03	Prospectivity report excerpt; extract B-2/B-3/B-4 narrative evidence and data gaps.
63	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessmentMap_1-50000_v01_raw-scan.jpg	03	Prospectivity assessment map; digitize B-3 Tol Khar/G-1 and priority areas.
64	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_1-50000_v01_raw-scan.jpg	03	Source materials map; digitize routes, stations, samples, trenches, pits and sections.
65	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_Legend_1-50000_v01_raw-scan.jpg	03	Source materials legend; build route/observation/sample/work-type domains.
66	05_Geology_Mineral_Prospectivity	2013_MN_L47-XIX_GoldOccurrenceDescriptions_XIX-74-A_4186_v01_raw.docx	03	Gold occurrence descriptions; extract Au occurrence coordinates, grades and geological descriptions.

67	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralOccurrenceAndMineralizedPointRegister_7255_v01_raw-scan.pdf	03	Mineral occurrence/mineralized point register; extract/cross-check occurrence attributes and P3 notes.
68	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralizedPointRegister_7255_v01_raw-table.xlsx	03	Mineralized point table; import cleaned coordinates/attributes to occurrence database.
69	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_Legend_RawScan_2020_v01.jpg	03	Regional metallogenic legend; build belt/ore-formation/commodity domain values.
70	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_RawScan_2020_v01.jpg	03	1:500k metallogenic map; use as regional metallogenic context only.
71	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book01_ProjectBook13_1M500K_RawScan_2021_v01.pdf	03	Metallogenic report book 01; extract text evidence for ore formation and regional context.
72	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book04_ProjectBook16_1M500K_RawScan_2021_v01.pdf	03	Metallogenic report book 04; extract text evidence for ore formation and regional context.
73	07_Basemap_Sentinel2_ASTER	2005-09-05_MN_ASTER-L1B_MultispectralImagery_00409052005043503_v01_raw.hdf	02	ASTER HDF raw product; import/extract bands and derive alteration/lithology indices.
74	07_Basemap_Sentinel2_ASTER	2025-05-28_MN_T46TGS_GeoreferencedSatelliteRaster_v01_raw.tif	02	Received satellite raster; QA/QC CRS/extent, subset/reproject and derive support products.
75	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_GoogleMaps_BasemapImagery_RGB_2p4m_WGS84_Raw_v01.tif	02	Google Maps RGB basemap; QA/QC/reproject for field planning and visual reference.
76	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_HighResolution_RGB_SurfaceBasemap_GoogleMaps_EPSG3857_0p15m_Raw_v01.tif	02	High-resolution surface basemap; reproject/clip for access, outcrop and field planning support.
77	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_GeologicalInterpretation_RGB_B12-B08-B03_10m_UTM46N_ReceivedRaw_v01.tif	02	Sentinel-2 geology composite; reproject to EPSG:32647 and use as lithology/alteration support.
78	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_LithologyIndex_B11B12_B08B11_B04B03_10m_UTM46N_ReceivedRaw_v01.tif	02	Sentinel-2 lithology index product; reproject and use as support evidence only.

1A.3 Mandatory rule for every phase

- Phase бүрийн аргачлалд input-ийг “geology files” эсвэл “remote sensing files” гэж дангаар бичихгүй; Section 1A.2 дахь raw input № болон exact filename-ийг заавал ашиглана.
- Output layer/report/table бүрт source_input_no, source_raw_filename, source_group, source_phase, processing_version, confidence, limitation талбар/мөр хадгална.
- Historical scanned map-derived evidence нь validation_status = Historical only байхаас field/lab confirmed evidence-тэй холигдохгүй.
- Remote sensing, DEM, KOMPSAT, Sentinel, ASTER, drone/LiDAR, pXRF output нь хүдэржилтийн баталгаа биш; support evidence гэж тэмдэглэнэ.

1B. Phase-wise exact raw input file processing and output matrix — v6 update

Зорилго. Энэ v6 нэмэлт нь 78 raw input file бүрийг аль phase-д ашиглах, яг ямар нэртэй input файл болох, ямар software/program-аар ямар боловсруулалт хийх, ямар нэртэй output file гаргахыг нэг бүрчлэн заана. Raw file-г overwrite хийхгүй; бүх ажил processing copy дээр хийгдэнэ. Output filename нь standard deliverable naming бөгөөд бодит боловсруулалтын үед version дугаарыг v01/v02... гэж өсгөнө.

1B.1 Phase тус бүрийн raw input file assignment summary

Workflow phase	Raw input file numbers	Main software/program	Main output package
00_Raw_Files_Archive	1-78	File system, checksum utility, Excel	Master inventory, checksum register, source data README
01_Phase_1_Data_Audit_and_Master_GIS_Setup	1-78 metadata; primary spatial boundary input №8; all raster/scan/doc sidecar QA	QGIS, GDAL/OGR, Excel	Master_GIS_Database.gpkg; Master_QGIS_Project.qgz; CRS/Georef QAQC Log; Data Confidence Ranking
02_Phase_2_Remote_Sensing_Preprocessing	9-46, 73-78	SNAP 13.0.0, QGIS, GDAL, ILWIS 3.6.8, Global Mapper	Processed DEM/Sentinel/ASTER/KOMPSAT outputs, QAQC logs, remote sensing support layers
03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	1-7, 53-72 plus outputs from Phase 1/2 and boundary №8	QGIS, QGIS Georeferencer, Excel/Word evidence registers	Geological evidence layers, metallogenic context, occurrence database, Preliminary Deposit Model.docx
04_Phase_4_Preliminary_Prospect_Delineation_and_Ranking	Derived from raw №1-8, 47-78 and Phase 2/3 outputs	QGIS, Excel scoring matrix	Prospect polygons, ranking table, Go/No-Go desktop matrix
05_Phase_5_DJI_Matrice_400_Drone_LiDAR_Photogrammetry_Survey	Derived A/B prospect outputs from Phase 4; DEM/access from №9-22 and basemap №75-78	DJI Matrice 400, Zenmuse P1/L2/L3, processing software, QGIS	Drone flight plan, orthomosaic, LiDAR point cloud, DTM/DSM, interpretation layers
06_Phase_6_Recon_Mapping_and_Portable_XRF_Field_Screening	Field planning from №51, №52, №60, №63, №64 and Phase 4/5 outputs	QField/QGIS, Olympus Vanta M, Bruker Titan S1	Recon traverse, field observations, pXRF register and QAQC report
07_Phase_7_Rock_Chip_Channel_and_Verification_Sampling	Field-confirmed targets derived from №55, №60, №63, №64, №66-68 and Phase 6 outputs	QField/QGIS, GPS/GNSS, lab submission templates	Rock chip/channel register, lab submission, assay import template
08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	47-52 plus target/geology outputs from №55, №60, №63, №64	QGIS, DEM drainage tools, pXRF, lab workflow	Orientation soil plan, stream/heavy mineral follow-up plan
09_Phase_9_Systematic_Soil_Grid_and_Laboratory_QAQC	Derived from Phase 8; geologic/target controls from №55, №60, №63, №64	QGIS grid design, pXRF, laboratory assay	Soil grid plan, sample points, QAQC report, soil assay results
10_Phase_10_Integrated_Interpretation_and_Final_Target_Ranking	All validated raw-derived outputs from №1-78 plus lab/field/drone results	QGIS, Excel/statistical validation, Word report	Integrated interpretation report, final target polygons, target description sheets
11_Phase_11_Follow_Up_Trench_Geophysics_and_Scout_Drill_Planning	Final A/B targets; DEM/access from №9-22; geology/structure from №53-68; RS from №73-78	QGIS, trench/geophysics/drilling planning templates	Follow-up work plan, trench/geophysics lines, scout drilling proposal, collar table

1B.2 Detailed 78 raw input file → software → processing → output matrix

Phase 1 inputs: boundary, full metadata audit and Master GIS setup

№	Evidence group	Exact raw input filename	Primary workflow phase	Software / program	Processing action	Expected output filename(s)
8	01_Tectonic_Terrane_KMZ	MN_BuduunKhad_L23222_LicenseBoundary_WGS84_v01_raw.kmz	01_Phase_1_Data_Audit_and_Master_GIS_Setup	QGIS, GDAL/OG	KMZ/KML polygon import; CRS	XV023222_Buduunkhad_L23222_LicenseBoundary_EPSG32647_v01.gpkg; XV023222_Buduunkhad_LicenseBoundary_QAQC_Register.xlsx;

				R, GeoPackage	шалгах; EPSG:4326-аас EPSG:32647 руу export; topology/area/perimeter QA/QC; master project-ийн үндсэн boundary layer болгоно.	XV023222_Buduunkhad_Project_Buffer_500m_1km_5km_10km_20km_EPSG32647.gpkg

Phase 2 inputs: DEM, ALOS-PALSAR, KOMPSAT-2, ASTER, Sentinel and basemap processing

№	Evidence group	Exact raw input filename	Primary workflow phase	Software / program	Processing action	Expected output filename(s)
9	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_DEM_1arcsec_WGS84_v01_raw.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, Global Mapper	DEM CRS/resolution/NoData/extent шалгах; license+buffer subset; EPSG:32647 reproject; hillshade/slope/aspect/contour/drainage derivative гаргана.	XV023222_Buduunkhad_ASTERGDEM_DEM_EPSG32647_v01.tif; XV023222_Buduunkhad_ASTERGDEM_Hillshade_Slope_Aspect_Drainage_EPSG32647_v01.gpkg/tif; XV023222_Buduunkhad_DEM_QAQC_Log.xlsx
10	02_DEM_ALOS_ASTERGDEM	ASTER-GDEM-v3_N45E096_NumObservations_1arcsec_WGS84_v01_raw.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	ASTER GDEM observation-count raster-г DEM quality mask болгон шалгах; low reliability pixel flag; DEM QA/QC-д холбох.	XV023222_Buduunkhad_ASTERGDEM_NumObservations_QA_Mask_EPSG32647_v01.tif; XV023222_Buduunkhad_DEM_Quality_Assessment.xlsx
11	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tfw	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, Global Mapper	Parent DEM-тэй хамтад нь bundle болгон хадгална; CRS/pixel size/extent/NoData шалгаж EPSG:32647 DEM derivative-ийн үндсэн эх болгон ашиглана.	XV023222_Buduunkhad_ALOS_PALSAR_DEM_12p5m_EPSG32647_v01.tif; XV023222_Buduunkhad_ALOS_PALSAR_Terrain_Derivatives_EPSG32647_v01.gpkg/tif; sidecar_bundle_QAQC entry (world file)
12	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, Global Mapper	ALOS-PALSAR 12.5 m DEM-ийг CRS/pixel/extent шалгаж subset/reproject; terrain derivative, drainage, access/safety, drone/trench planning-д ашиглана.	XV023222_Buduunkhad_ALOS_PALSAR_DEM_12p5m_EPSG32647_v01.tif; XV023222_Buduunkhad_ALOS_PALSAR_Terrain_Derivatives_EPSG32647_v01.gpkg/tif; sidecar_bundle_QAQC entry (12.5 m DEM raster)
13	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.aux.xml	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, Global Mapper	Parent DEM-тэй хамтад нь bundle болгон хадгална; CRS/pixel size/extent/NoData шалгаж EPSG:32647 DEM derivative-ийн үндсэн эх болгон ашиглана.	XV023222_Buduunkhad_ALOS_PALSAR_DEM_12p5m_EPSG32647_v01.tif; XV023222_Buduunkhad_ALOS_PALSAR_Terrain_Derivatives_EPSG32647_v01.gpkg/tif; sidecar_bundle_QAQC entry (aux/statistics sidecar)
14	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_DEM_12p5m_UTM47N_Raw_v01.tif.ovr	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, Global Mapper	Parent DEM-тэй хамтад нь bundle болгон хадгална; CRS/pixel size/extent/NoData шалгаж EPSG:32647 DEM derivative-ийн үндсэн эх болгон ашиглана.	XV023222_Buduunkhad_ALOS_PALSAR_DEM_12p5m_EPSG32647_v01.tif; XV023222_Buduunkhad_ALOS_PALSAR_Terrain_Derivatives_EPSG32647_v01.gpkg/tif; sidecar_bundle_QAQC entry (overview pyramid sidecar)

15	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tfw	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_hillshade_world_file_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
16	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_hillshade_raster_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
17	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.aux.xml	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_hillshade_aux_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
18	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_Hillshade_12p5m_UTM47N_Derived_v01.tif.ovr	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_hillshade_overview_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
19	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tfw	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_slope_world_file_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
20	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_slope_raster_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
21	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.aux.xml	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах; terrain/access/lineament interpretation-д reference layer болгоно.	XV023222_Buduunkhad_ALOS_PALSAR_slope_aux_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx
22	02_DEM_ALOS_ASTERGDEM	XV023222_Buduunkhad_ALOS-PALSAR_SlopeDeg_12p5m_UTM47N_Derived_v01.tif.ovr	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	Derived raster/sidecar QA; parent raster-тэй хамт bundle audit; EPSG:32647 alignment шалгах;	XV023222_Buduunkhad_ALOS_PALSAR_slope_overview_QAQC_EPSG32647_v01.tif/register; XV023222_Buduunkhad_Terrain_Derivatives_Index.xlsx

					terrain/access/lineament interpretation-д reference layer болгоно.	
23	03_KOMPSAT2_MSC_L1G	KOMPSAT EULA Form_3.1.pdf	00_Raw_Files_Archive / 02_Phase_2_Remote_Sensing_Preprocessing	PDF reader, Word/Excel register	Data license/provenance бүртгэл; usage restriction, source note, acquisition info-г Source Data README-д оруулна.	XV023222_Buduunkhad_KOMPSAT2_Data_License_Provenance_Register.xlsx; XV023222_Buduunkhad_Source_Data_Readme.docx
24	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	PAN raster: PAN orthorectification, terrain correction, pan-sharpen base, high-resolution lineament/outcrop/access interpretation; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
25	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.txt	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	PAN metadata: PAN metadata extraction: acquisition, GSD, projection/source details; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
26	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.rpc	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	PAN RPC: RPC terrain correction/orthorectification support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
27	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344PN00_1G.eph	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	PAN ephemeris: orbit/geometry support archive; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
28	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Green band: MS band alignment and RGB composite component; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit,	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx

					orthorectification, band stack, true/false color and NDVI support products.	
29	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.txt	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Green metadata: band metadata extraction; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
30	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.rpc	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Green RPC: MS orthorectification support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
31	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M1N00G_1G.eph	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Green ephemeris: orbit/geometry support archive; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
32	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Blue band: MS band alignment and RGB component; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
33	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.txt	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Blue metadata: band metadata extraction; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
34	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.rpc	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Blue RPC: MS orthorectification support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
35	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M2N00B_1G.eph	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8,	Blue ephemeris: orbit/geometry support archive;	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap

				RPC/orthorectification workflow	PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
36	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	NIR band: false color/NDVI/vegetation mask component; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
37	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.txt	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	NIR metadata: band metadata extraction; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
38	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.rpc	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	NIR RPC: MS orthorectification support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
39	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M3N00N_1G.eph	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	NIR ephemeris: orbit/geometry support archive; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
40	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Red band: RGB/NDVI component, iron-stained surface visual interpretation; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
41	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.txt	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Red metadata: band metadata extraction; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit,	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Inter

					orthorectification, band stack, true/false color and NDVI support products.	pretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
42	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.rpc	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Red RPC: MS orthorectification support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
43	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344M4N00R_1G.eph	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	Red ephemeris: orbit/geometry support archive; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
44	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jpg	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	browse image: coverage preview/catalog thumbnail only; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
45	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_br.jgw	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	browse world file: browse georeference sidecar support; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
46	03_KOMPSAT2_MSC_L1G	MSC_111127030410_28454_08621344N00_1G_tn.jpg	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, ILWIS 3.6.8, RPC/orthorectification workflow	thumbnail: inventory thumbnail only; PAN/MS bundle alignment, RPC/EPH/TXT metadata audit, orthorectification, band stack, true/false color and NDVI support products.	XV023222_Buduunkhad_KOMPSAT2_PAN_MS_Orthorectified_Bundle_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap_EPSG32647_v01.tif; XV023222_Buduunkhad_KOMPSAT2_NDVI_Lineament_Outcrop_Interpretation_EPSG32647_v01.gpkg; XV023222_Buduunkhad_KOMPSAT2_QAQC_Register.xlsx
73	07_Basemap_Sentinel2_ASTER	2005-09-05_MN_ASTER-L1B_MultispectralImagery_00409052005043503_v01_raw.hdf	02_Phase_2_Remote_Sensing_Preprocessing	SNAP 13.0.0, ILWIS/ASTER workflow v5, QGIS/GDAL	ASTER HDF import; VNIR/SWIR/TIR band extraction where available; projection to UTM47/EPSG32647; alteration/lithology indices; score/class/binary mask separation.	XV023222_Buduunkhad_ASTER_BandStack_EPSG32647_v01.tif; XV023222_Buduunkhad_ASTER_score_porphyry_alteration_raw_v01.tif; XV023222_Buduunkhad_ASTER_porphyry_potential_class_v01.tif; XV023222_Buduunkhad_ASTER_porphyry_final_target_binary_mask_v01.tif; ASTER_QAQC_Log.xlsx

74	07_Basemap_Sentinel2_ASTER	2025-05-28_MN_T46TGS_GeoreferencedSatelliteRaster_v01_raw.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL, SNAP if source L1C/L2A needed	CRS/extent/band check; subset/reproject to EPSG:32647; base satellite raster QA; derivative composite support.	XV023222_Buduunkhad_20250528_T46TGS_GeoreferencedSatelliteRaster_EPSG32647_v01.tif; RemoteSensing_QAQC_Register.xlsx
75	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_GoogleMaps_BasemapImagery_RGB_2p4m_WGS84_Raw_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	WGS84 basemap CRS check; reproject to EPSG:32647; visual reference layer for field route/access/outcrop interpretation.	XV023222_Buduunkhad_GoogleMaps_Basemap_RGB_2p4m_EPSG32647_v01.tif; Basemap_Overlay_QAQC_Log.xlsx
76	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_HighResolution_RGB_SurfaceBasemap_GoogleMaps_EPSG3857_0p15m_Raw_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	QGIS, GDAL	EPSG:3857 high-resolution basemap reproject; tile/clip to license+buffer; use for detailed outcrop/disturbance/access interpretation.	XV023222_Buduunkhad_HighResolution_RGB_SurfaceBasemap_GoogleMaps_0p15m_EPSG32647_v01.tif; HighRes_Basemap_QAQC_Log.xlsx
77	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_GeologicalInterpretation_RGB_B12-B08-B03_10m_UTM46N_ReceivedRaw_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	SNAP 13.0.0, QGIS, GDAL	Sentinel-2 geology RGB B12-B08-B03 status check; UTM46N to EPSG32647 reproject; subset; lithology/structure visual interpretation support.	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_GeologicalInterpretation_RGB_B12-B08-B03_10m_EPSG32647_v01.tif; Sentinel2_Geology_RGB_QAQC_Log.xlsx
78	07_Basemap_Sentinel2_ASTER	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_LithologyIndex_B11B12_B08B11_B04B03_10m_UTM46N_ReceivedRaw_v01.tif	02_Phase_2_Remote_Sensing_Preprocessing	SNAP 13.0.0, QGIS, GDAL	Sentinel-2 lithology index raster QA; UTM46N to EPSG32647; mask/noise check; geology/alteration support layer, ore proof биш.	XV023222_Buduunkhad_Sentinel2_T46TGS_20250528_LithologyIndex_B11B12_B08B11_B04B03_10m_EPSG32647_v01.tif; Sentinel2_LithologyIndex_QAQC_Log.xlsx

Phase 3 / 03A inputs: tectonic, geology, mineral occurrence, prospectivity and metallogenic synthesis

№	Evidence group	Exact raw input filename	Primary workflow phase	Software / program	Processing action	Expected output filename(s)
1	01_Tectonic_Terrane_KMZ	Geological and Tectonic Characteristics of the Lake Terrane, Mongolia.png	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Lake Terrane-ийн геологи/тектоникийн context зураг/эх мэдээлэл-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
2	01_Tectonic_Terrane_KMZ	Mongolia_Tectonic_Terrane_Map_Project_Area_Lake_Island_Arc_Terrane.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	теслийн талбай Lake island-arc terrane-тэй холбогдох regional context-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg

					terrane/metallogenic context болгон digitize/attribute coding хийнэ.	
3	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Central_Mongolian_Massif_and_Daagandel_Tectonic_Zone_Page11.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Даагандэлийн бүс ба Төв Монголын массивын тайлбар-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
4	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part1_Page08.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Нуурын аккрецийн мегабүсийн тайлбар-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
5	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part2_Page09.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Хурайн/Баатархайрхан/Улааншандын бүсийн тайлбар-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
6	01_Tectonic_Terrane_KMZ	MUGZ500_Geomed2013_Explanatory_Text_Nuur_Accretionary_Megazone_Part3_Page10.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Ханхөхий/Хантайширийн бүсийн тайлбар-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
7	01_Tectonic_Terrane_KMZ	Regional_Tectonic_Subdivision_Map_of_Mongolia_Tumurtogoo_2017_Buduunkhad_Project_in_Ulaanshand_Zone.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS Layout, Excel/Word evidence register	Улааншандын бүсийн regional tectonic context-ийг source confidence-тэй бүртгэж, шаардлагатай бол georeference/overlay хийж terrane/metallogenic context болгон digitize/attribute coding хийнэ.	XV023222_Buduunkhad_Tectonic_Terrane_Context_Register.xlsx; XV023222_Buduunkhad_Tectonic_Terrane_Context_Map_EPSG32647.pdf; XV023222_Buduunkhad_Tectonic_Context_Layers.gpkg
53	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_1-200000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer,	Georeference; regional geology unit/fault/intrusive/cont	XV023222_Buduunkhad_1987_L47-XIX_GeologicalMap_1-200K_Georeferenced_EPSG32647_v01.tif;

				QGIS digitizing	act vectorization; scale flag = regional.	geology_units_200k_polygons/structures_faults_lines_EPSG32647_v01.gpkg
54	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_Legend_1-200000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS/Image viewer, Excel lookup	Regional geology legend dictionary; lithology/age/intrusive/structure domain values.	XV023222_Buduunkhad_1987_GeologicalMap_Legend_Symbol_Dictionary_v01.xlsx
55	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; detailed lithology, intrusive, contact, fault, vein, alteration/section digitizing; target-scale geology layer.	XV023222_Buduunkhad_2013_L47-74-A_GeologicalMap_1-50K_Georeferenced_EPSG32647_v01.tif; geology_units_50k_polygons/structures_faults_lines/intrusive_contacts_lines/dyke_vein_lines_EPSG32647_v01.gpkg
56	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_Legend_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS/Image viewer, Excel lookup	Detailed geology legend dictionary; stratigraphic_unit, vein_type, alteration, lithology domains.	XV023222_Buduunkhad_2013_GeologicalMap_Legend_Symbol_Dictionary_v01.xlsx
57	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_1-200000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; regional occurrence/resource/anomaly/ore field digitizing; commodity coding.	XV023222_Buduunkhad_1987_L47-XIX_MineralResourcesMap_1-200K_Georeferenced_EPSG32647_v01.tif; mineral_occurrences_points/mineralized_zones_polygons/ore_field_prospect_polygons_EPSG32647_v01.gpkg
58	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_Legend_1-200000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS/Image viewer, Excel lookup	Mineral resources legend dictionary; commodity, occurrence type, ore field type domains.	XV023222_Buduunkhad_1987_MineralResources_Legend_Symbol_Dictionary_v01.xlsx
59	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralDistributionPatternMap_1-100000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; ore district/node/mineral distribution context digitizing; Phase 3 deposit model context.	XV023222_Buduunkhad_2013_L47-73-74_MineralDistributionPatternMap_1-100K_Georeferenced_EPSG32647_v01.tif; ore_district_node_context_EPSG32647_v01.gpkg
60	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralOccurrenceMap_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; Au-Cu/Cu/Mo/As/Zn occurrence points and target features digitizing; relation to geology/source materials.	XV023222_Buduunkhad_2013_L47-74-A_MineralOccurrenceMap_1-50K_Georeferenced_EPSG32647_v01.tif; mineral_occurrences_points_EPSG32647_v01.gpkg; XV023222_Buduunkhad_Mineral_Occurrences_Register.xlsx
61	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MetallogenicSchemeAndMetallogenogram_1-400000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Regional metallogenic scheme georeference/context digitizing; ore formation and age/element relation register.	XV023222_Buduunkhad_2013_MetallogenicSchemeMetallogenogram_1-400K_Georeferenced_EPSG32647_v01.tif; metallogenic_context_layers_EPSG32647_v01.gpkg
62	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessment_ReportExcerpt_B3-TolKhar_v01_raw-photo.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	Image/PDF viewer, Word/Excel evidence register	Report excerpt transcription/summary; B-2/B-3/B-4 area evidence, limitation, recommended follow-up coding.	XV023222_Buduunkhad_2013_Prospectivity_ReportExcerpt_Evidence_Register_v01.xlsx; report_evidence_summary_for_Preliminary_Deposit_Model.docx
63	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessmentMap_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; Б-3 Толь хяр, Г-1 and other prospectivity polygons digitizing; priority and evidence_basis attributes.	XV023222_Buduunkhad_2013_L47-74-A_ProspectivityAssessmentMap_1-50K_Georeferenced_EPSG32647_v01.tif; prospectivity_target_zones_polygons_EPSG32647_v01.gpkg; XV023222_Buduunkhad_Prospectivity_Target_Register.xlsx
64	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference; route, observation, sample, trench/pit/shaft/channel, section line digitizing for QField/field validation.	XV023222_Buduunkhad_2013_L47-74-A_SourceMaterialsMap_1-50K_Georeferenced_EPSG32647_v01.tif; source_material_observation_points/source_material_route_lines/source_material_trench_pit_points_EPSG32647_v01.gpkg

65	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_Legend_1-50000_v01_raw-scan.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS/Image viewer, Excel lookup	Source materials legend dictionary; observation_type, sample_type, work_type domains; QField form lookup.	XV023222_Buduunkhad_SourceMaterials_Legend_Symbol_Dictionary_v01.xlsx; QField_Lookups_Domains.xlsx
66	05_Geology_Mineral_Prospectivity	2013_MN_L47-XIX_GoldOccurrenceDescriptions_XIX-74-A_4186_v01_raw.docx	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	Microsoft Word/LibreOffice, Excel, QGIS	Gold occurrence descriptions extraction; coordinates/grade/lithology/structure; occurrence register and model evidence coding.	XV023222_Buduunkhad_4186_GoldOccurrence_Descriptions_Extracted_Register_v01.xlsx; XV023222_Buduunkhad_4186_GoldOccurrence_Points_EPSG32647_v01.gpkg; preliminary_deposit_model_evidence_table.xlsx
67	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralOccurrenceAndMineralizedPointRegister_7255_v01_raw-scan.pdf	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	PDF reader, Excel, QGIS	PDF register extraction; occurrence/mineralized point attributes; coordinate validation and cross-reference with map 60/68.	XV023222_Buduunkhad_7255_MineralOccurrence_MineralizedPoint_Register_Extracted_v01.xlsx; XV023222_Buduunkhad_7255_MineralizedPoint_Layers_EPSG32647_v01.gpkg; extraction_QAQC_log.xlsx
68	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralizedPointRegister_7255_v01_raw-table.xlsx	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	Excel/LibreOffice Calc, QGIS	XLSX table cleaning; coordinate conversion; element/commodity standardization; GIS point layer and deposit model evidence.	XV023222_Buduunkhad_7255_MineralizedPoint_Clean_Register_v01.xlsx; XV023222_Buduunkhad_7255_MineralizedPoint_Points_EPSG32647_v01.gpkg; XV023222_Buduunkhad_Occurrence_CrossReference_7255_4186.xlsx
69	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_Legend_RawScan_2020_v01.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS/Image viewer, Excel lookup	1:500k metallogenic legend symbol dictionary; ore formation/commodity/metallogenic unit domains.	XV023222_Buduunkhad_L47B_RegionalMetallogenic_Legend_Dictionary_v01.xlsx
70	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_RawScan_2020_v01.jpg	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	QGIS Georeferencer, QGIS digitizing	Georeference/context overlay; metallogenic belt/zone/ore district/node digitizing; context_only flag.	XV023222_Buduunkhad_2020_L47B_Talshand_RegionalMetallogenicMap_1-500K_Georeferenced_EPSG32647_v01.tif; metallogenic_zones_polygons_EPSG32647_v01.gpkg; regional_metallogenic_context_map.pdf
71	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book01_ProjectBook13_1M500K_RawScan_2021_v01.pdf	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	PDF reader, Word/Excel evidence register	Regional report text/table extraction; ore formation/metallogenic context notes for deposit model.	XV023222_Buduunkhad_RegionalMetallogenic_Report_Book01_Evidence_Register_v01.xlsx; metallogenic_context_summary.docx
72	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book04_ProjectBook16_1M500K_RawScan_2021_v01.pdf	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 03A	PDF reader, Word/Excel evidence register	Regional report text/table extraction; cross-reference with map 70/71; context evidence and limitations.	XV023222_Buduunkhad_RegionalMetallogenic_Report_Book04_Evidence_Register_v01.xlsx; metallogenic_context_cross_reference.xlsx

Phase 6 and Phase 8 field/historical geochemistry inputs

№	Evidence group	Exact raw input filename	Primary workflow phase	Software / program	Processing action	Expected output filename(s)
47	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_1-200000_v01_raw-scan.jpg	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	QGIS Georeferencer, QGIS digitizing, DEM drainage analysis	Georeference; heavy mineral sample/anomaly/indicator contour digitizing; DEM drainage catchment overlay; upstream source check planning.	XV023222_Buduunkhad_1987_L47-XIX_HeavyMineralSamplingResultsMap_1-200K_Georeferenced_EPSG32647_v01.tif; XV023222_Buduunkhad_HeavyMineral_Anomaly_Layers_EPSG32647_v01.gpkg; XV023222_Buduunkhad_HeavyMineral_FollowUp_Plan.xlsx/pdf

48	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	QGIS/Image viewer, Excel lookup table	Legend symbol dictionary үүсгэх; mineral_indicator/anomaly_class/sample_symbol domain values; main map digitizing attribute control.	XV023222_Buduunkhad_HeavyMineral_Symbol_Dictionary_v01.xlsx; QGIS lookup/domain table in gpkg
49	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	QGIS/Image viewer, Excel lookup table	Stream sediment legend dictionary; element suite/anomaly level/contour type domain; map interpretation QA.	XV023222_Buduunkhad_StreamSediment_Symbol_Dictionary_v01.xlsx; QGIS lookup/domain table in gpkg
50	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Polyelement_1-200000_v01_raw-scan.jpg	08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check	QGIS Georeferencer, QGIS digitizing, DEM drainage analysis	Georeference; Cu-Pb-Zn-Ag-As-Bi-W-Sn-Mo-Mn-Ba-F anomaly contour/vector digitizing; drainage source direction and orientation sampling plan.	XV023222_Buduunkhad_1987_L47-XIX_StreamSedimentPolyelementMap_1-200K_Georeferenced_EPSG32647_v01.tif; XV023222_Buduunkhad_StreamSediment_Anomaly_Layers_EPSG32647_v01.gpkg; XV023222_Buduunkhad_StreamSediment_FollowUp_Plan.xlsx/pdf
51	04_HeavyMineral_StreamSediment_Field	2011_MN_Namalzakh_L47-74-A_FieldRouteNotebook_Routes14-15_Obs1076-1090_1-50000_v01_raw-scan.pdf	06_Phase_6_Recon_Mapping_and_Portable_XRF_Field_Screening / 08_Phase_8	PDF reader, QGIS, Excel register	Route 14-15 and observations 1076-1090 extraction; coordinates/route/sketch/observation attributes; QField recon and sampling planning.	XV023222_Buduunkhad_2011_FieldRouteNotebook_Routes14-15_Observation_Register_v01.xlsx; XV023222_Buduunkhad_Field_Route_Observation_Layers_EPSG32647_v01.gpkg; XV023222_Buduunkhad_QField_Recon_Input_Package_v01
52	04_HeavyMineral_StreamSediment_Field	XV023222_Buduunkhad_Report7255_FieldObservation_StationDescription_Table_WGS84_LegacyRaw_v01.pdf	06_Phase_6_Recon_Mapping_and_Portable_XRF_Field_Screening / 03_Phase_3	PDF reader, Excel, QGIS	Station description table extraction; WGS84 coordinates validation; lithology/structure/mineralization attribute coding; Master GIS observation layer.	XV023222_Buduunkhad_Report7255_FieldObservation_Station_Register_v01.xlsx; XV023222_Buduunkhad_Report7255_FieldObservation_Points_EPSG32647_v01.gpkg; XV023222_Buduunkhad_FieldObservation_QAQC_Log.xlsx

Тайлбар. Phase 4-11 нь raw file-г шууд дахин боловсруулахаас илүүтэй Phase 1-3/03A болон Phase 8-аас гарсан QA/QC хийгдсэн derivative output-уудыг ашиглана. Гэсэн ч тэдгээр derivative output бүрийн эх raw input №-г дээрх matrix-д хадгалсан тул final target sheet, sample register, drill plan бүрт source_raw_input_no/source_file/source_phase талбаруудыг заавал үлдээнэ.

1B.3 Required source-traceability fields for every output

Output field	Required value	Purpose
source_raw_input_no	1-78 дугаар	Output бүрийг аль raw input-оос үүссэнтэй audit trail-аар холбох
source_raw_filename	Exact raw filename	Файлын нэр өөрчлөгдөх/холилдох эрсдэлийг хаах
processing_phase	00/01/02/03/03A/04...	Аль workflow шатанд боловсруулсан болохыг тодруулах
processing_software	QGIS / SNAP / GDAL / Excel / QField / etc.	Дахин боловсруулахад reproducibility хангах
processing_action	Georeference / reproject / digitize / extract / score / QAQC	Юу хийснийг тодорхой бичих
output_filename	Standardized output file name	Deliverable package-д нэг мөр нэршилтэй хадгалах
qaqc_status	Draft / Checked / Approved / Rejected	Decision-grade биш output-г буруу ашиглахаас хамгаалах
validation_status	Historical only / Field checked / Sampled / Lab confirmed	Historical evidence ба баталгаажсан evidence-г салгах

v6 implementation note

Энэ v6 хувилбар нь 78 raw input file бүрийг phase-wise байдлаар exact filename, software, processing action, expected output filename-тай холбосон тул цаашид QGIS/Excel/QField/Drone/Lab workflow-д source-traceability хадгалахад ашиглана. Бүх output нь preliminary/support evidence бөгөөд field validation, laboratory assay, QA/QC review хийгдэх хүртэл mineralization proof эсвэл resource/reserve estimate биш.

2. Integrated 00-99 phase workflow

```
00_Raw_Files_Archive
  -> 01_Phase_1_Data_Audit_and_Master_GIS_Setup
  -> 02_Phase_2_Remote_Sensing_Preprocessing
  -> 03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis
  -> 04_Phase_4_Preliminary_Prospect_Delineation_and_Ranking
  -> 05_Phase_5_DJI_Matrice_400_Drone_LiDAR_Photogrammetry_Survey
  -> 06_Phase_6_Recon_Mapping_and_Portable_XRF_Field_Screening
  -> 07_Phase_7_Rock_Chip_Channel_and_Verification_Sampling
  -> 08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check
  -> 09_Phase_9_Systematic_Soil_Grid_and_Laboratory_QAQC
  -> 10_Phase_10_Integrated_Interpretation_and_Final_Target_Ranking
  -> 11_Phase_11_Follow_Up_Trench_Geophysics_and_Scout_Drill_Planning
  -> 99_Final_Deliverables
```

00. Raw Files Archive

Subsection	Methodology detail
Зорилго	Raw data-г өөрчлөхгүй архивлах, integrity/metadata/source хяналт тогтоох.
Input files	Direct raw input files: №1-78 бүх raw input file. Яг filename, evidence group, primary phase, methodology action-ийг Section 1A-д бүрэн заасан. Sidecar files (.tfw, .aux.xml, .ovr, .rpc, .eph, .txt) parent raster/image-ээс салгахгүй.
Software / equipment	File system, checksum utility, inventory workbook.

Processing folder structure

```
00_Raw_Files_Archive/
├── 01_Tectonic_Terrane_KMZ
├── 02_DEM_ALOS_ASTERGDEM
├── 03_KOMPSAT2_MSC_L1G
├── 04_HeavyMineral_StreamSediment_Field
├── 05_Geology_Mineral_Prospectivity
├── 06_Regional_Metallogenic_L47B
└── 07_Basemap_Sentinel2_ASTER
```

Step-by-step methodology

1. 78 raw input файлыг evidence group-ийн дагуу raw archive хавтас руу байршуулна.
2. Original filename, standardized filename, file type, source note, owner/responsible person, read status, processing copy status бүртгэнэ.
3. SHA-256 checksum үүсгэж integrity log-д бичнэ.
4. Sidecar файлуудыг parent file-аас салгахгүй: .tfw/.aux.xml/.ovr/.rpc/.eph/.txt metadata нь тухайн raster/image bundle-ийн хэсэг.
5. Raw file дээр rename/overwrite/clip/reproject хийхгүй; working copy-г дараагийн phase-д хуулна.

QA/QC check

QA/QC item	Acceptance note
Checksum match	Recorded in phase QA/QC log; reviewer/date/decision required.
Raw overwrite хийгдээгүй	Recorded in phase QA/QC log; reviewer/date/decision required.
Sidecar completeness	Recorded in phase QA/QC log; reviewer/date/decision required.
Source note and owner registered	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_78Input_Master_Inventory.xlsx
- XV-023222_Buduunkhad_Raw_Data_Integrity_Log.xlsx
- XV-023222_Buduunkhad_Source_Data_Readme.docx
- SHA-256_Checksum_Register.csv

Decision gate / next phase condition

- All files archived, checksum complete, processing copies available.

01. Phase 1 — Data Audit and Master GIS Setup

Subsection	Methodology detail
Зорилго	78 input file-ийг GIS-ready болгох, EPSG:32647 master database үүсгэх.
Input files	Direct raw input files: №1-78 бүх raw input file. Үүнээс №8 license boundary нь master boundary layer; №9-78 raster/scan/table/PDF/DOCX бүрийн metadata, CRS, sidecar, confidence, working-copy status шалгана. Section 1A-г primary input checklist болгон ашиглана.
Software / equipment	QGIS, GeoPackage, spreadsheet register.

Processing folder structure

```
01_Phase_1_Data_Audit_and_Master_GIS_Setup/
├── 01_File_Inventory
├── 02_Metadata_Check
├── 03_CRS_Check
├── 04_Raster_Scan_Georeference_QAQC
├── 05_KMZ_KML_to_GPKG
├── 06_Master_GeoPackage_Schema
```

```
├── 07_Data_Confidence_Ranking
└── 08_Master_QGIS_Project_Setup
```

Step-by-step methodology

6. QGIS project үүсгэнэ: XV-023222_Buduunkhad_Master_QGIS_Project.qgz.
7. Project CRS-г WGS 84 / UTM Zone 47N, EPSG:32647 болгож тохируулна.
8. MN_BuduunKhad_L23222_LicenseBoundary_WGS84_v01_raw.kmz-г import хийж GeoPackage layer болгон хадгална.
9. Raster бүрийн CRS, resolution, extent, NoData, band count, pixel alignment, sidecar availability-г шалгана.
10. Scan map бүрт georeference QA/QC: GCP count, residual, map scale, grid/tick consistency, reviewer/date/decision бүртгэнэ.
11. Master GeoPackage schema үүсгэнэ: license_boundary, geology_units_polygon, faults_structures_line, intrusive_contacts_line, mineral_occurrences_point, stream_sediment_anomaly_polygon, heavy_mineral_anomaly_polygon, lineament_interpretation_line, preliminary_prospect_polygon, target_polygon, field_observation_point, sample_point, pXRF_reading_table.
12. Data confidence ranking: High / Medium / Low / Needs verification өгнө.

QA/QC check

QA/QC item	Acceptance note
EPSG:32647 project CRS	Recorded in phase QA/QC log; reviewer/date/decision required.
KMZ boundary topology valid	Recorded in phase QA/QC log; reviewer/date/decision required.
Raster CRS/resolution/extent/nodata/band count checked	Recorded in phase QA/QC log; reviewer/date/decision required.
Scan georeference residual and confidence logged	Recorded in phase QA/QC log; reviewer/date/decision required.
Master GPKG schema created	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Master_GIS_Database.gpkg
- XV-023222_Buduunkhad_Master_QGIS_Project.qgz
- XV-023222_Buduunkhad_CRS_Georeference_QAQC_Log.xlsx
- XV-023222_Buduunkhad_Data_Confidence_Ranking.xlsx

Decision gate / next phase condition

- Master GIS project opens without missing layers; critical data confidence recorded.

Нэмэлт дэд аргачлал: Historical scanned map файлуудыг QGIS дээр inventory -> georeference -> vector digitizing -> register -> QA/QC -> confidence ranking -> Master GIS handover дараалллаар боловсруулах нарийвчилсан SOP-ийг энэ баримт бичгийн Appendix E хэсэгт бүрэн оруулав. Энэ дэд аргачлал нь Phase 1 Data Audit, Phase 3 Geological/Metallogenic Synthesis, Phase 4 Prospect Ranking, Phase 6 Recon Mapping, Phase 7 Sampling, Phase 8/9 Soil/Stream/Heavy Mineral planning-д шууд handover хийх зориулалттай.

02. Phase 2 — Remote Sensing Preprocessing

Subsection	Methodology detail
Зорилго	Sentinel, ASTER, KOMPSAT, DEM өгөгдлийг QA/QC-тэй interpretation-ready support layer болгох.

Input files	Direct raw input files: №9-22 DEM/ALOS/ASTERGDEM; №23-46 KOMPSAT-2 PAN/MS/RPC/EPH/metadata/browse; №73 ASTER HDF; №74-78 Sentinel/Google/basemap rasters. Exact filename бүрийг Section 1A болон Section 1 register-ээс авна.
Software / equipment	SNAP 13.0.0, ASTER workflow v5/ILWIS, ILWIS 3.6.8 or QGIS, Global Mapper 24.0/QGIS.

Processing folder structure

```
02_Phase_2_Remote_Sensing_Preprocessing/
├── 01_Sentinel2_SNAP13
├── 02_ASTER_Workflow_v5
├── 03_KOMPSAT2_ILWIS368_QGIS
├── 04_ALOS_ASTERGDEM_GlobalMapper_QGIS
├── 05_RemoteSensing_QAQC
└── 06_Export_EPSG32647
```

Step-by-step methodology

13. Sentinel-2 raw/received raster status шалгана; L1C бол SNAP 13.0.0 Sen2Cor ашиглан L2A болгоно; received derivative бол metadata-г бүртгэнэ.
14. Subset: license boundary + 500 m эсвэл 1 km buffer.
15. Resample: all relevant bands to 10 m grid; pixel alignment шалгана.
16. Cloud/shadow/snow/water/vegetation mask үүсгэж alteration/lithology interpretation-д noise орохоос сэргийлнэ.
17. Sentinel RGB/index: Natural RGB, SWIR-NIR-Red, geology composite, lithology ratio/index, NDVI, NDWI, iron oxide, ferric, clay/SWIR, ferrous, brightness index.
18. ASTER workflow v5: HDF import, band extraction, UTM47 project grid, b*_project band-аас index тооцох; haze/edge filter-ийг ratio calculation-д ашиглахгүй.
19. ASTER outputs тусад нь хадгална: raw Float32 score, 1/2/3 class, 0/1 binary mask.
20. KOMPSAT-2: PAN/MS, RPC, EPH, TXT metadata-г хамт хадгалж orthorectification/band alignment/pan-sharpen/NDVI/lineament-outcrop basemap гаргана.
21. ALOS-PALSAR/ASTER GDEM: hillshade, slope, aspect, contour, drainage, watershed, terrain ruggedness, curvature, access/safety derivatives гаргана.

QA/QC check

QA/QC item	Acceptance note
Cloud/shadow/vegetation mask applied where relevant	Recorded in phase QA/QC log; reviewer/date/decision required.
ASTER raw score/class/binary mask separated	Recorded in phase QA/QC log; reviewer/date/decision required.
KOMPSAT PAN/MS alignment checked	Recorded in phase QA/QC log; reviewer/date/decision required.
DEM derivatives visually and spatially checked	Recorded in phase QA/QC log; reviewer/date/decision required.
No remote sensing output treated as ore proof	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Sentinel2_Processed_Products.tif/gpkg
- XV-023222_Buduunkhad_ASTER_score_porphyry_alteration_raw_v1.tif
- XV-023222_Buduunkhad_ASTER_porphyry_potential_class_v1.tif
- XV-023222_Buduunkhad_ASTER_porphyry_final_target_binary_mask_v1.tif
- XV-023222_Buduunkhad_KOMPSAT2_Pansharpened_Orthobasemap.tif
- XV-023222_Buduunkhad_Terrain_Derivatives.gpkg
- XV-023222_Buduunkhad_RemoteSensing_QAQC_Report.docx

Decision gate / next phase condition

- Remote sensing derivatives passed QA/QC and are ready as support evidence only.

03. Phase 3 — Geological, Metallogenic and CMCS Synthesis

Subsection	Methodology detail
Зорилго	Геологи, металлогени, орд-илрэлийн context, deposit model candidate-уудыг нэгтгэх.
Input files	Direct raw input files: №1-7 tectonic/terrane context; №8 license boundary for overlay/buffer; №53-68 geology, mineral resources, mineral occurrence, prospectivity, source material and occurrence/register files; №69-72 regional metallogenic map/report files. Exact filename бүр Section 1A-д заасан.
Software / equipment	QGIS, CMCS/MRPAM query, geological interpretation register.

Processing folder structure

```
03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis/
├── 01_Tectonic_Terrane_Context
├── 02_1M500K_Metallogenic_Overlay
├── 03_1M200K_Regional_Geology_Mineral
├── 04_1M50K_Geology_Occurrence_Prospectivity
├── 05_CMCS_MRPAM_Buffer_Check_5km_10km_20km
├── 06_Deposit_Model_Screening
├── 07_Preliminary_Deposit_Model_Preparation
├── 08_Model_Evidence_Scoring_and_Ranking
└── 09_Geological_Evidence_GPKG
```

Step-by-step methodology

22. 1:500,000 metallogenic map, 1:200,000 regional geology/mineral resources, 1:50,000 geology/mineral occurrence/source/prospectivity map-уудыг нэг CRS-д давхарлана.
23. CMCS/MRPAM nearest deposit/occurrence check-ийг L23222 boundary-гаас 5 km, 10 km, 20 km buffer-аар хийж register үүсгэнэ.
24. Tectonic terrane болон Ulaanshand Zone / Nuur Accretionary Megazone context-ийг deposit model screening-д ашиглана.
25. Mineral occurrence, mineralized point, source material, field notebook, gold occurrence description зэргийг GIS database-д холбож coordinate/attribute validation хийнэ.
26. CMCS evidence нь contextual support болохоос тухайн license дотор хүдэржилт байгаа эсэхийг батлахгүй гэж тэмдэглэнэ.

QA/QC check

QA/QC item	Acceptance note
Map scale limitations recorded	Recorded in phase QA/QC log; reviewer/date/decision required.
CMCS context not used as direct proof	Recorded in phase QA/QC log; reviewer/date/decision required.
Occurrence coordinates validated	Recorded in phase QA/QC log; reviewer/date/decision required.
Deposit model evidence/missing evidence table completed	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_CMCS_Nearest_Deposit_Register.xlsx
- XV-023222_Buduunkhad_Regional_Metallogenic_Context_Map.pdf
- XV-023222_Buduunkhad_Preliminary_Deposit_Model.docx
- XV-023222_Buduunkhad_Geological_Evidence_Layers.gpkg

Decision gate / next phase condition

- Deposit model candidates and contextual evidence are documented with limitations.

03A. Preliminary Deposit Model Preparation — Phase 3 доторх дэд workflow

Энэ хэсэг нь XV-023222_Buduunkhad_Preliminary_Deposit_Model.docx файлыг бэлтгэх аргачлалыг үндсэн workflow-ийн Phase 3 дотор байрлуулсан болно. Энэ нь Appendix биш; Phase 3-ийн Geological, Metallogenic and CMCS Synthesis ажлын заавал хийх дэд ажил бөгөөд Phase 4 preliminary prospect ranking, Phase 10 final target ranking руу шууд handover хийнэ.

Энэхүү дэд workflow нь ордын төрлийн урьдчилсан концепцийн загвар гаргах аргачлал юм. Satellite, ASTER, KOMPSAT-2, DEM, Drone/LiDAR болон pXRF output нь хүдэржилтийн баталгаа биш; эдгээр нь target generation, field validation, sampling prioritization-д ашиглах support evidence. Эцсийн confidence нь хээрийн шалгалт, дээжлэлт, лабораторийн шинжилгээ, structural/geological evidence, шаардлагатай бол trench/geophysics/scout drilling-аар баталгаажна.

03A.1 Зорилго ба гарах баримт бичиг

XV-023222_Buduunkhad_Preliminary_Deposit_Model.docx нь Будүүн хад талбайд боломжит ордын төрлүүдийг урьдчилсан байдлаар ялгаж, ямар evidence дээр үндэслэж байгаа, ямар evidence дутуу байгаа, дараагийн ямар field/lab validation хийх шаардлагатайг тодорхойлох концепцийн загварын баримт бичиг байна.

Энэ файлыг таамгаар бичихгүй. 78 input workflow-ийн Phase 3 — Geological, Metallogenic and CMCS Synthesis шатны үр дүнд, Phase 1-ийн Master GIS database болон historical scanned map vectorization output дээр тулгуурлан гаргана.

Талбар	Утга
Project area	Buduunkhad / XV-023222 / L23222
Standard CRS	WGS 84 / UTM Zone 47N, EPSG:32647
Document to be prepared	XV-023222_Buduunkhad_Preliminary_Deposit_Model.docx
Document type	Preliminary conceptual deposit model methodology; final resource/reserve estimate биш
Source workflow basis	78 Inputs v3/v4 Enhanced workflow + Historical Scanned Maps QGIS Vectorization v02 Detailed
Workflow location	03_Phase_3_Geological_Metallogenic_and_CMCS_Synthesis / 07_Preliminary_Deposit_Model_Preparation

03A.2 Ашиглах input evidence

Input evidence	Ашиглах мэдээлэл	Deposit model-д өгөх үүрэг
Геологийн суурь зураг	1:50,000, 1:200,000 geology map, legend, lithology, intrusive, structure, fault, vein, alteration.	Host rock, intrusive contact, structural control, local/regional geology.
Ашигт малтмалын илрэл, эрдэсжсэн цэг	Mineral occurrence map, mineral resources map, 7255 register, 4186 gold occurrence description, mineralized point table.	Au, Cu, Mo, As, Zn, Pb, W, Sn, Bi element association болон historical occurrence evidence.

Металлогений зураг, тайлан	1:500,000 metallogenic map, L47B regional metallogenic report, metallogenic scheme/metallogenogram.	Metallogenic belt, ore formation, ore district/node context. Local target boundary биш.
Historical scanned map vector output	Georeferenced raster, geology, structure, occurrence, stream sediment, heavy mineral, prospectivity target vector layers.	Historical evidence database. validation_status = Historical only гэж хадгална.
Remote sensing support	Sentinel, ASTER, KOMPSAT, DEM, ALOS-PALSAR derivative layers.	Alteration, lithology, lineament, terrain, drainage, exposure support. Ore proof биш.

03A.3 Ажлын үндсэн дараалал

Алхам 1 — Evidence layer-үүдийг Master GIS дээр нэгтгэх

QGIS дээр license_boundary, geology_units_50k/200k, structures_faults_lines, intrusive_contacts_lines, mineral_occurrences_points, mineralized_zones, stream_sediment_anomaly, heavy_mineral_anomaly, prospectivity_target_zones, metallogenic_zones, Sentinel/ASTER/KOMPSAT/DEM support layer-үүдийг EPSG:32647 CRS-тэй нэг project-д давхарлана.

Алхам 2 — Historical map-уудаас ордын төрлийн evidence ялгах

Historical scanned map vectorization workflow-ийн дагуу map type бүрээс host geology, structure, occurrence, geochemical anomaly, heavy mineral, prospectivity zone, metallogenic context ялгана. Historical vector data-г field/lab confirmed data-тай холихгүй.

Алхам 3 — Deposit model candidate-уудыг тодорхойлох

Au-Cu hydrothermal vein, intrusion-related Cu-Au-Mo, skarn/contact metasomatic, polymetallic vein, VMS-type sulphide possibility, heavy mineral/placer indicator гэсэн candidate model бүрийг тусад нь шалгана.

Алхам 4 — Supporting evidence / missing evidence / validation work хүснэгт гаргах

Ордын төрөл бүр дээр одоо байгаа evidence, дутуу evidence, баталгаажуулах ажлыг хүснэгтээр үнэлнэ.

Алхам 5 — Evidence weight ашиглаж preliminary ranking хийх

Deposit model тус бүрт 100 онооны matrix-ээр оноо өгч High priority model / Moderate priority model / Low conceptual model / Insufficient evidence гэж ангилна.

Map type	Deposit model-д авах мэдээлэл
Geological map	Host rock, intrusive contact, volcanic/intrusive package, structure.
Mineral occurrence map	Au-Cu, Cu, Mo, As, Zn зэрэг илрэл.
Mineral resources map	Regional occurrence, ore field, anomaly.
Stream sediment map	Cu-Pb-Zn-Ag-As-Bi-W-Sn-Mo-Mn-Ba-F anomaly.
Heavy mineral map	Au, W, Sn, Ti, Cr, magnetite зэрэг indicator.
Prospectivity map	Б-3 Толь хяр, Г-1 зэрэг хэтийн төлөвтэй хэсэг.
Metallogenic map	Ore formation, metallogenic belt, ore district context.

03A.4 Deposit model candidate screening

Candidate deposit model	Юуг шалгах вэ?	Дутуу evidence / эрсдэл	Recommended validation work
Au-Cu hydrothermal vein	Quartz vein, pyrite, chalcopyrite, malachite, Au-Cu-Ag-As-Bi, shear/fault corridor, lineament intersection.	Au pXRF unreliable; судлын continuity, width, grade тодорхойгүй байж болно.	Recon mapping, rock chip/channel, lab Au fire assay + multi-element, LiDAR/structural mapping.

Intrusion-related Cu-Au-Mo	Diorite/granodiorite/gabbrodiorite contact, Cu-Mo-Bi-As, stockwork/quartz veinlets, ASTER/Sentinel alteration support.	Porphyry-style alteration zoning and sulphide system not confirmed.	ASTER validation, intrusive phase mapping, soil grid, IP/magnetic, trench/channel.
Skarn/contact metasomatic Cu-Au-W-Bi	Intrusive-carbonate contact, epidote/garnet/magnetite/skarn minerals, W-Bi-Cu-Au association.	Carbonate host and skarn mineral assemblage unclear.	Detailed contact mapping, pXRF W/Bi/Cu screening, petrography, magnetic/IP support.
Polymetallic vein	Pb-Zn-Cu-Ag-As association, vein/shear structures, gossan/iron oxide, historical occurrence overlap.	Depth continuity and grade continuity unknown.	Rock chip/channel, soil grid, structural mapping, lab Pb-Zn-Ag multi-element.
VMS-type sulphide possibility	Volcanic-sedimentary package, Cu-Zn-Pb-Ba-Fe-Mn, stratiform sulphide or gossan, regional arc/oceanic context.	Stratigraphic control and massive sulphide textures not confirmed.	Detailed stratigraphic mapping, geochemistry, IP, magnetic, targeted trenching.
Heavy mineral / placer indicator	Historical shlich/heavy mineral anomaly, drainage concentration, Au/W/Sn/Ti/Cr indicators.	Source may be transported/secondary; bedrock source not proven.	Drainage follow-up, upstream sampling, heavy mineral panning, geomorphology and bedrock checking.

03A.5 Evidence weight ба preliminary ranking

Шалгуур	Оноо	Тайлбар
Favorable geology / host lithology	20	Host rock, intrusive/contact, volcanic-sedimentary or carbonate contact setting.
Intrusive/contact/structure control	15	Fault, shear, vein trend, lineament intersection, intrusive contact.
Known mineral occurrence	15	Historical Au-Cu/Cu/Mo/Pb-Zn/W-Sn occurrence or mineralized point.
Historical geochemistry / shlich / stream sediment	15	Cu-Au-Ag-Mo-Bi-As-Pb-Zn-W-Sn anomaly overlap, drainage source consistency.
Metallogenic context	10	Relevant metallogenic belt, ore formation, ore district/node context.
ASTER/Sentinel alteration support	10	Clay/sericite/silica/ferric/chlorite/carbonate indicators and lithology contrast.
Field mapping / pXRF support	10	Malachite, pyrite, epidote, quartz vein, elevated Cu/Pb/Zn/As/Mo/W/Sn; Au pXRF not decision-grade.
Access / workability	5	Field access, slope, drone/sampling/trenching feasibility.
Нийт	100	Deposit model тус бүрээр оноо өгнө.

Confidence class	Score range	Meaning
High priority model	>=70	Олон evidence давхцаж байгаа, field/lab follow-up priority.
Moderate priority model	50-69	Боломжтой боловч нэмэлт field/lab validation шаардлагатай.
Low / conceptual model	30-49	Contextual эсвэл дутуу evidence ихтэй.
Insufficient evidence	<30	Одоогийн өгөгдлөөр deposit model гэж дэмжихэд хангалтгүй.

Rank	Deposit model	Preliminary confidence	Why
1	Au-Cu hydrothermal vein	High / Moderate	Quartz vein, Au-Cu occurrence, structure support байвал өндөр оноо авна.
2	Intrusion-related Cu-Au-Mo	Moderate	Intrusive contact + Cu-Mo-Bi-As possibility + alteration support шалгана.
3	Skarn/contact metasomatic	Moderate / Low	Contact evidence байгаа ч carbonate/skarn mineral баталгаажуулах шаардлагатай.

4	Polymetallic vein	Moderate / Low	Pb-Zn-Cu-Ag-As association байгаа эсэхийг field/lab-аар шалгана.
5	VMS possibility	Conceptual	Regional volcanic context байж болох ч direct stratiform sulphide evidence дутуу.
6	Heavy mineral / placer	Contextual	Drainage/shlich evidence байж болох ч bedrock source тодорхойгүй.

03A.6 XV-023222_Buduunkhad_Preliminary_Deposit_Model.docx санал болгох бүтэц

- Title page: XV-023222 / L23222 Buduunkhad Preliminary Deposit Model; subtitle: Geological, Metallogenic, Historical Geochemistry, Remote Sensing and Occurrence Evidence-Based Conceptual Deposit Model.
- Methodology note: Final resource model биш, preliminary conceptual model гэдгийг тайлбарлана.
- Input data basis: 78 raw input-ийн аль evidence group-ээс ямар мэдээлэл авсныг хүснэгтээр оруулна.
- Regional geological setting: Ulaanshand Zone, Nuur Accretionary Megazone, Lake island-arc terrane context.
- Local geology and structural control: 1:50k, 1:200k geology, intrusive, fault, contact, vein, alteration.
- Mineral occurrence and geochemical evidence: Au-Cu, Cu, Mo, As, Zn, Pb, W, Sn, Bi, stream sediment, heavy mineral evidence.
- Remote sensing and terrain support: Sentinel, ASTER, KOMPSAT, ALOS/DEM support; ore proof биш гэдгийг заавал тэмдэглэнэ.
- Deposit model candidate screening: 6 candidate model-ийг supporting evidence / missing evidence / recommended validation work хүснэгтээр үнэлнэ.
- Preliminary model ranking: Хамгийн боломжтой ордын төрлийг evidence score-оор эрэмбэлнэ.
- Recommended validation work: Historical map QA/QC, recon mapping, pXRF, sampling, orientation soil, drainage follow-up, lab assay, IP/magnetic/trench/scout drilling.

03A.7 Богино ажлын checklist

- 78 input-ийн геологи, илрэл, геохими, металлогени, remote sensing evidence-г Master GIS-д нэгтгэнэ.
- Historical scanned map-уудыг georeference + vectorize хийж confidence ranking өгнө.
- Ордын боломжит төрлүүдийг Au-Cu vein, intrusion-related Cu-Au-Mo, skarn, polymetallic vein, VMS, placer/heavy mineral гэж ангилна.
- Төрөл бүрээр байгаа evidence, дутуу evidence, баталгаажуулах ажлыг хүснэгтээр үнэлнэ.
- Хамгийн боломжтой ордын төрлийг preliminary ranking-ээр гаргана.
- Энэ нь final дүгнэлт биш, field validation + lab assay дараа шинэчлэгдэх conceptual model гэж тэмдэглэнэ.

03A.8 Phase 3 QA/QC notes for deposit model preparation

- Raw data-г засварлахгүй; processing copy дээр ажиллана.
- Final deliverables-ийн CRS нь EPSG:32647; native/raw CRS-г metadata-д хадгална.
- Historical scanned map-derived vector data нь validation_status = Historical only гэж хадгалагдана.
- Regional 1:400k/1:500k metallogenic layer-ийг local target boundary мэт ашиглахгүй.
- ASTER final binary mask, Sentinel alteration ratio, KOMPSAT visual lineament, drone interpretation нь хүдэржилтийн баталгаа биш.
- pXRF нь lab assay-г орлохгүй; Au-ийн pXRF response-ийг decision-grade гэж үзэхгүй.
- CMCS/MRPAM nearest deposit нь contextual evidence бөгөөд тухайн license дотор хүдэржилт байгаа эсэхийг шууд батлахгүй.
- Final target confidence нь хээрийн шалгалт, дээжлэлт, laboratory assay, structural/geological evidence, шаардлагатай бол trench/geophysics/scout drilling-аар баталгаажна.

Methodology guide only — not mineralization proof or resource estimate

03A.9 Handover from Phase 3 to Phase 4 and Phase 10

03A дэд workflow-ийн үр дүн нь Phase 4-ийн Preliminary Prospect Delineation and Ranking-д deposit model evidence score, missing evidence, validation priority хэлбэрээр орно. Мөн Phase 10-д final target ranking хийх үед field/lab result-аар шинэчлэгдсэн conceptual model болгон дахин шалгана.

Handover item	Phase 4-д ашиглах байдал	Phase 10-д ашиглах байдал
Deposit model candidate table	Prospect polygon бүрийн model-fit шалгуур болно.	Final target sheet-д model-fit confidence болгон шинэчилнэ.
Evidence weight score	Preliminary A/B/C/D ranking-ийн geology/model component болно.	Assay/field validation-аар re-score хийнэ.
Missing evidence / risk	Field validation, drone, recon, sampling priority гаргана.	Go/Conditional Go/No-Go decision-д data gap/risk болно.
Recommended validation work	Phase 5-9 ажлын дараалал, sample plan, orientation survey-г чиглүүлнэ.	Trench/geophysics/scout drilling criteria-д шилжинэ.

04. Phase 4 — Preliminary Prospect Delineation and Ranking

Subsection	Methodology detail
Зорилго	Desktop evidence-үүдийг 100 онооны scoring matrix-аар preliminary prospect болгох.
Input files	Input files: Phase 1-3 processed outputs + source raw inputs №1-78 as traceable evidence basis. Key direct evidence sources: №47-52 geochemistry/field observations, №53-68 geology/occurrence/prospectivity/source materials, №69-72 metallogenic context, №9-46 and №73-78 terrain/remote sensing support, №8 license boundary.
Software / equipment	QGIS, scoring spreadsheet, prospect register.

Processing folder structure

```
04_Phase_4_Preliminary_Prospect_Delineation_and_Ranking/
├── 01_Evidence_Overlay
├── 02_Prospect_Polygon_Delineation
├── 03_Scoring_Matrix
├── 04_Confidence_DataGap_NextAction
└── 05_A_B_C_D_Field_Priority
```

Step-by-step methodology

27. Evidence overlay үүсгэнэ: geology + occurrence + stream/heavy mineral + ASTER/Sentinel + KOMPSAT lineament/outcrop + DEM terrain + CMCS context.
28. Prospect polygon бүрт evidence score, confidence flag, limitation/data gap, field access, safety risk, next action бүртгэнэ.
29. A/B/C/D preliminary target class олгоно: A >=75, B=55-74, C=35-54, D<35.
30. Field-ready A/B prospect-уудыг drone survey болон recon mapping-д шилжүүлнэ.

Phase 3-ийн 03A Preliminary Deposit Model Preparation-аас гарсан deposit model score, missing evidence, recommended validation work-ийг prospect scoring matrix-д заавал холбож оруулна. Prospect polygon бүрт dominant_deposit_model, model_confidence, missing_model_evidence, validation_priority талбар нэмнэ.

QA/QC check

QA/QC item	Acceptance note
100-point matrix calculated	Recorded in phase QA/QC log; reviewer/date/decision required.
Confidence/data gap/next action fields filled	Recorded in phase QA/QC log; reviewer/date/decision required.

A/B/C/D class reviewed	Recorded in phase QA/QC log; reviewer/date/decision required.
Field access and safety checked	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Preliminary_Prospect_Ranking_Map.pdf
- XV-023222_Buduunkhad_Prospect_Polygons.gpkg
- XV-023222_Buduunkhad_Prospect_Ranking_Table.xlsx
- XV-023222_Buduunkhad_Go_NoGo_Desktop_Decision_Matrix.xlsx

Decision gate / next phase condition

- A/B prospects selected for drone and recon; C/D retained with data gaps.

05. Phase 5 — DJI Matrice 400 Drone LiDAR Photogrammetry Survey

Subsection	Methodology detail
Зорилго	Priority prospect дээр orthomosaic/LiDAR/terrain/structure field planning base авах.
Input files	Input files: Phase 4 A/B prospect polygons plus direct planning support raw inputs №8 license boundary, №9-22 DEM/slope/hillshade, №24-46 KOMPSAT basemap/lineament support, №75-78 high-resolution/Sentinel/basemap rasters. Exact filename list is in Section 1A.
Software / equipment	4 x DJI Matrice 400, Zenmuse P1, Zenmuse L2, Zenmuse L3, GNSS/RTK/PPK, processing software.

Processing folder structure

```
05_Phase_5_DJI_Matrice_400_Drone_LiDAR_Photogrammetry_Survey/
├── 01_Flight_Block_Design
├── 02_GCP_Checkpoint_RTK_PPK
├── 03_Zenmuse_P1_Photogrammetry
├── 04_Zenmuse_L2_LiDAR
├── 05_Zenmuse_L3_Detailed_LiDAR
├── 06_Raw_Backup_Flight_Log
├── 07_Processing_Orthomosaic_PointCloud_DTM_DSM
└── 08_Drone_QAQC_Interpretation
```

Step-by-step methodology

31. 4 ширхэг DJI Matrice 400-г parallel survey team болгон ашиглана: P1 orthomosaic, L2 terrain/structure, L3 detailed LiDAR, 4-р дрон backup/parallel block.
32. Flight block design: target boundary + buffer, terrain/slope/access, take-off/landing/emergency area, no-fly/safety restriction.
33. GCP/checkpoint, RTK/PPK, overlap, flight altitude, wind/weather/sun angle, battery rotation, raw backup, flight log бүртгэнэ.
34. Zenmuse P1: high-resolution orthomosaic, oblique photo, outcrop mapping base.
35. Zenmuse L2: DTM/DSM, terrain, drainage, slope, structural lineament.
36. Zenmuse L3: detailed LiDAR, micro-topography, fault/contact/vein corridor.
37. Output-уудыг field traverse, sample point, trench/drill pad planning-д ашиглана.

QA/QC check

QA/QC item	Acceptance note
GCP/checkpoint accuracy reviewed	Recorded in phase QA/QC log; reviewer/date/decision required.
Flight log complete	Recorded in phase QA/QC log; reviewer/date/decision required.
Overlap/altitude/weather recorded	Recorded in phase QA/QC log; reviewer/date/decision required.
Raw photo/LiDAR backed up	Recorded in phase QA/QC log; reviewer/date/decision required.
DTM/DSM/orthomosaic checked	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Drone_Flight_Plan.pdf
- XV-023222_Buduunkhad_Drone_Orthomosaic_P1.tif
- XV-023222_Buduunkhad_Drone_LiDAR_PointCloud.laz
- XV-023222_Buduunkhad_Drone_DTM_DSM.tif
- XV-023222_Buduunkhad_Drone_Structure_Outcrop_Interpretation.gpkg

Decision gate / next phase condition

- Orthomosaic/LiDAR products meet mapping scale and field planning requirements.

06. Phase 6 — Recon Mapping and Portable XRF Field Screening

Subsection	Methodology detail
Зорилго	Priority target-ийг газар дээр шалгаж pXRF vectoring хийх.
Input files	Input files: Phase 4 target polygons + Phase 5 drone outputs + direct validation support raw inputs №55-56 detailed geology/legend, №60 mineral occurrence map, №63 prospectivity map, №64-65 source materials map/legend, №66-68 occurrence/register files, №9-22 terrain, №75-78 basemaps, №8 boundary.
Software / equipment	QField/QGIS forms, Olympus Vanta M, Bruker Titan S1, GPS/GNSS, camera.

Processing folder structure

```
06_Phase_6_Recon_Mapping_and_Portable_XRF_Field_Screening/
├── 01_Traverse_Planning
├── 02_Field_Mapping_Forms
├── 03_pXRF_VantaM_Primary
├── 04_pXRF_TitanS1_Duplicate_Check
├── 05_pXRF_QAQC_CRM_Blank_Duplicate
├── 06_Field_Database
└── 07_Recon_Report
```

Step-by-step methodology

38. A/B targets дээр traverse төлөвлөж lithology, alteration, mineralization, vein, structure, weathering, exposure, access, safety бүртгэнэ.
39. Olympus Vanta M-г primary screening, Bruker Titan S1-г duplicate/cross-check байдлаар ашиглана.
40. Өдөр бүр warm-up, calibration check, CRM, blank, duplicate, check sample уншуулна.

41. pXRF бүртгэлд sample ID, GPS coordinate, lithology, alteration, mineralization, instrument model/serial, operator, mode, reading time, moisture/surface condition, Cu/Pb/Zn/As/Mo/W/Sn/Mn/Fe/S зэрэг element орно.
42. Au-ийн pXRF response-ийг decision-grade evidence гэж үзэхгүй; lab assay шаардлагатай.
43. pXRF-lab correlation sheet-д element бүрээр reliability flag өгнө.

QA/QC check

QA/QC item	Acceptance note
CRM/blank/duplicate/check sample daily	Recorded in phase QA/QC log; reviewer/date/decision required.
Vanta M vs Titan S1 duplicate comparison	Recorded in phase QA/QC log; reviewer/date/decision required.
Au not used as pXRF decision-grade	Recorded in phase QA/QC log; reviewer/date/decision required.
Moisture/surface condition recorded	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Recon_Traverse_Lines.gpkg
- XV-023222_Buduunkhad_Field_Observation_Points.gpkg
- XV-023222_Buduunkhad_pXRF_Field_Screening_Register.xlsx
- XV-023222_Buduunkhad_pXRF_QAQC_Report.docx
- XV-023222_Buduunkhad_Recon_Mapping_Report.docx

Decision gate / next phase condition

- Field evidence and pXRF vectoring justify sampling or downgrade target.

07. Phase 7 — Rock Chip, Channel and Verification Sampling

Subsection	Methodology detail
Зорилго	Field-confirmed mineralization/alteration/structure дээр лабораторийн дээж авах.
Input files	Input files: Phase 6 recon/pXRF outputs + direct historical evidence raw inputs №52 field observation table, №55-56 detailed geology/legend, №60 mineral occurrence map, №63 prospectivity map, №64-65 source materials map/legend, №66-68 occurrence/register files, №9-22 terrain and №75-78 basemaps.
Software / equipment	Field sampling kit, GPS/GNSS, pXRF support, sample bags/tags, chain-of-custody forms.

Processing folder structure

```
07_Phase_7_Rock_Chip_Channel_and_Verification_Sampling/
├── 01_Sample_Planning
├── 02_RockChip_Channel_Float_Registers
├── 03_QAQC_Insertion
├── 04_Chain_of_Custody
├── 05_Lab_Submission
└── 06_Assay_Import_Preparation
```

Step-by-step methodology

44. Recon/pXRF-ээр баталгаажсан quartz vein, gossan, malachite/sulphide, intrusive contact, shear/fault, altered lithology дээр дээж авна.
45. Sample type: representative rock chip, selective rock chip, float, channel, verification sample.
46. Sample ID convention: BUD-RC-001, BUD-CH-001, BUD-SOIL-001, BUD-STR-001, BUD-HM-001, BUD-QC-STD-001, BUD-QC-BLK-001, BUD-QC-DUP-001.
47. Sample register: coordinate, photo, drone tile, pXRF reading, lithology, alteration, mineralization, structure, width/strike/dip, sample mass, collector, date/time.
48. QA/QC: CRM/standard, blank, duplicate, field duplicate, lab duplicate, pulp duplicate; chain-of-custody ба lab submission template заавал бүрдүүлнэ.

QA/QC check

QA/QC item	Acceptance note
Sample ID unique	Recorded in phase QA/QC log; reviewer/date/decision required.
Coordinates/photo/chain-of-custody complete	Recorded in phase QA/QC log; reviewer/date/decision required.
QA/QC insertion complete	Recorded in phase QA/QC log; reviewer/date/decision required.
Lab submission consistent with register	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Rock_Chip_Sampling_Plan.pdf
- XV-023222_Buduunkhad_Rock_Chip_Sample_Register.xlsx
- XV-023222_Buduunkhad_Rock_Chip_QAQC_Plan.xlsx
- XV-023222_Buduunkhad_Lab_Submission_RockChip.xlsx
- XV-023222_Buduunkhad_Assay_Import_Template.xlsx

Decision gate / next phase condition

- Lab submission complete and QA/QC inserted; assay import template ready.

08. Phase 8 — Orientation Soil, Stream Sediment and Heavy Mineral Check

Subsection	Methodology detail
Зорилго	Systematic grid өмнө soil/drainage response аргачлал баталгаажуулах.
Input files	Direct raw input files: №47 HeavyMineralSamplingResultsMap, №48 HeavyMineral legend, №49 StreamSediment legend, №50 StreamSediment Polyelement map, №51 field route notebook, №52 field observation table; support inputs №9-22 DEM/drainage, №53-56 geology, №60 occurrence map, №63-64 prospectivity/source materials, №68 mineralized point table.
Software / equipment	Soil auger/shovel/sieve, GPS, pXRF, lab submission workflow.

Processing folder structure

```
08_Phase_8_Orientation_Soil_StreamSediment_and_HeavyMineral_Check/
├── 01_Orientation_Line_Design
├── 02_Depth_Horizon_Mesh_Test
├── 03_pXRF_Lab_Comparison
├── 04_StreamSediment_FollowUp
```

```
├── 05_HeavyMineral_FollowUp
└── 06_Recommended_Systematic_Method
```

Step-by-step methodology

49. Systematic grid шууд эхлүүлэхгүй; эхлээд orientation survey хийж horizon/depth/mesh/spacing response баталгаажуулна.
50. Depth test: 20 cm, 40 cm, 60-80 cm. Horizon: A/B/C/residual/transported.
51. Mesh/fraction test, soil texture, carbonate, clay, slope position, drainage/alluvial influence тэмдэглэнэ.
52. pXRF + lab comparison-аар ямар element suite, horizon, depth, spacing илүү anomaly өгч байгааг тодорхойлно.
53. Historical stream sediment/heavy mineral map-тай drainage catchment analysis хийж upstream source direction болон follow-up point төлөвлөнө.

QA/QC check

QA/QC item	Acceptance note
Depth/horizon/mesh comparison complete	Recorded in phase QA/QC log; reviewer/date/decision required.
Transported vs residual flag	Recorded in phase QA/QC log; reviewer/date/decision required.
pXRF-lab comparison done	Recorded in phase QA/QC log; reviewer/date/decision required.
Drainage source logic documented	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Orientation_Soil_Survey_Plan.pdf
- XV-023222_Buduunkhad_Orientation_Soil_Sample_Register.xlsx
- XV-023222_Buduunkhad_Orientation_Soil_pXRF_Lab_Comparison.xlsx
- XV-023222_Buduunkhad_StreamSediment_FollowUp_Plan.pdf
- XV-023222_Buduunkhad_HeavyMineral_FollowUp_Plan.pdf

Decision gate / next phase condition

- Best horizon/depth/mesh/spacing confirmed before systematic grid.

09. Phase 9 — Systematic Soil Grid and Laboratory QA/QC

Subsection	Methodology detail
Зорилго	Validated method-оор systematic soil geochemical coverage хийх.
Input files	Input files: Phase 8 orientation results + direct planning support raw inputs №8 boundary, №9-22 DEM/slope/drainage, №47-52 historical drainage/heavy mineral/field evidence, №55/60/63/64/68 local geology-occurrence-source evidence, №75-78 basemaps/Sentinel support. Exact filenames in Section 1A.
Software / equipment	QGIS grid design, field collection tools, pXRF, laboratory assay.

Processing folder structure

```
09_Phase_9_Systematic_Soil_Grid_and_Laboratory_QAQC/
├── 01_Grid_Design_200x50_100x25_50x25_25x10
├── 02_Field_Collection
```

```
├── 03_pXRF_Screening
├── 04_Lab_Submission_QAQC
├── 05_Assay_Validation
└── 06_Soil_Anomaly_Map
```

Step-by-step methodology

54. Orientation result-оор grid spacing сонгоно: Recon 200 m x 50 m, Target 100 m x 25 m, Infill 50 m x 25 m, Vein detail 25 m x 10 m.
55. Grid orientation нь geological strike, structural trend, drainage/slope-д нийцсэн байна.
56. pXRF realtime screening ашиглаж field vectoring хийж болох боловч final map lab assay дээр үндэслэнэ.
57. QA/QC insertion schedule: CRM, blank, duplicate, field duplicate, lab repeat, pulp duplicate.
58. Assay validation: unit conversion, detection limit, duplicate/CRM/blank performance, outlier check.

QA/QC check

QA/QC item	Acceptance note
Grid spacing justified by orientation results	Recorded in phase QA/QC log; reviewer/date/decision required.
QA/QC performance acceptable	Recorded in phase QA/QC log; reviewer/date/decision required.
Assay unit/detection limit validated	Recorded in phase QA/QC log; reviewer/date/decision required.
Anomaly continuity checked	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Systematic_Soil_Grid_Plan.pdf
- XV-023222_Buduunkhad_Soil_Sample_Points.gpkg
- XV-023222_Buduunkhad_Soil_Sample_Register.xlsx
- XV-023222_Buduunkhad_Soil_QAQC_Report.docx
- XV-023222_Buduunkhad_Soil_Assay_Results.xlsx

Decision gate / next phase condition

- Validated geochemical anomaly supports final target ranking.

10. Phase 10 — Integrated Interpretation and Final Target Ranking

Subsection	Methodology detail
Зорилго	Бүх evidence, assay QA/QC, remote/drone/field data-г final target decision болгон нэгтгэх.
Input files	Input files: All validated phase outputs + full traceable raw evidence basis №1-78. Final target sheets must reference exact source raw input filenames from Section 1A/Section 1, not only generic evidence group names.
Software / equipment	QGIS, spreadsheet/statistical validation, report templates.

Processing folder structure

```
10_Phase_10_Integrated_Interpretation_and_Final_Target_Ranking/
├── 01_Assay_Validation
├── 02_Evidence_Integration
├── 03_Target_Scoring
├── 04_Target_Description_Sheets
├── 05_Go_NoGo_Decision
└── 06_Final_Target_GIS_Map_Report
```

Step-by-step methodology

59. Geology, metallogeny, mineral occurrence, stream sediment, heavy mineral, Sentinel, ASTER, KOMPSAT, ALOS DEM, drone, LiDAR, field mapping, pXRF, rock chip/channel assay, soil assay, CMCS context-ийг нэгтгэнэ.
60. Assay validation: unit conversion, detection limits, duplicate check, CRM/blank performance, pXRF-lab correlation, outlier check.
61. ASTER/Sentinel support layers-ийг field/lab evidence-тэй давхарлаж зөвхөн validation-supported target polygon болгон засварлана.
62. Final target description sheet бүрт target ID, location, evidence summary, geology, structure, alteration, geochemistry, remote sensing, drone/LiDAR, sampling result, confidence, risk/data gap, recommended follow-up, Go/No-Go decision бичнэ.

QA/QC check

QA/QC item	Acceptance note
All evidence layers version-controlled	Recorded in phase QA/QC log; reviewer/date/decision required.
Assay QA/QC passed	Recorded in phase QA/QC log; reviewer/date/decision required.
pXRF-lab correlation documented	Recorded in phase QA/QC log; reviewer/date/decision required.
Target sheets complete	Recorded in phase QA/QC log; reviewer/date/decision required.
Go/No-Go decision recorded	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Integrated_Interpretation_Report.docx
- XV-023222_Buduunkhad_Integrated_Geology_Geochemistry_Alteration_Map.pdf
- XV-023222_Buduunkhad_Final_Target_Polygons.gpkg
- XV-023222_Buduunkhad_Final_Target_Ranking_Table.xlsx
- XV-023222_Buduunkhad_Target_Description_Sheets.docx

Decision gate / next phase condition

- Final A/B targets have sufficient evidence for trench/geophysics/scout drill planning.

11. Phase 11 — Follow-up Trench, Geophysics and Scout Drill Planning

Subsection	Methodology detail
Зорилго	Final A/B target дээр trench, geophysics, scout drilling төлөвлөх.

Input files	Input files: Phase 10 final A/B targets + direct planning support raw inputs №8 boundary, №9-22 DEM/slope/hillshade, №55 detailed geology, №60 occurrence map, №63 prospectivity map, №64 source materials map, №68 mineralized point table, №75-78 basemap/Sentinel rasters.
Software / equipment	QGIS, trench/geophysics planning tools, drilling design spreadsheet, HSE/budget templates.

Processing folder structure

```
11_Phase_11_Follow_Up_Trench_Geophysics_and_Scout_Drill_Planning/
├── 01_Trench_Channel_Planning
├── 02_IP_Resistivity_Planning
├── 03_Magnetic_Survey_Planning
├── 04_Infill_Soil_Planning
├── 05_Scout_Drill_Collar_Design
├── 06_HSE_Environment_Rehabilitation
└── 07_Budget_Permit_Schedule
```

Step-by-step methodology

63. Trench/channel хийх нөхцөл: surface mineralization + lab/pXRF response + accessible slope + geometry trace.
64. IP/resistivity хийх нөхцөл: disseminated sulphide/chargeability target эсвэл covered anomaly; line orientation нь structure/geology-г огтлох.
65. Magnetic survey хийх нөхцөл: intrusive/mafic/contact/structure ялгах шаардлагатай үед.
66. Infill soil хийх нөхцөл: open-ended soil anomaly, low spacing confidence, transported cover uncertainty.
67. Scout drilling minimum criteria: confirmed surface mineralization, lab assay support, structure confirmed, favorable geology/contact, target geometry estimated, access/HSE possible, trench/geophysics recommended or completed.
68. Drill collar table: collar ID, Easting/Northing, RL, azimuth, dip, depth, target, section line, access, pad, water, HSE, rehabilitation, budget.

QA/QC check

QA/QC item	Acceptance note
Minimum scout drill criteria met	Recorded in phase QA/QC log; reviewer/date/decision required.
HSE/access/rehabilitation reviewed	Recorded in phase QA/QC log; reviewer/date/decision required.
Drill collar geometry justified	Recorded in phase QA/QC log; reviewer/date/decision required.
Budget/permit/schedule template completed	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- XV-023222_Buduunkhad_Follow_Up_Work_Plan.pdf
- XV-023222_Buduunkhad_Proposed_Trench_Locations.gpkg
- XV-023222_Buduunkhad_Proposed_IP_Magnetic_Lines.gpkg
- XV-023222_Buduunkhad_Scout_Drilling_Proposal.docx
- XV-023222_Buduunkhad_Drill_Collar_Table.xlsx

Decision gate / next phase condition

- Scout drilling proceeds only if minimum criteria and HSE/access/permit conditions are met.

99. Final Deliverables

Subsection	Methodology detail
Зорилго	Бүх output-ийг стандарт folder package болгон бүрдүүлэх.
Input files	Input files: All phase outputs and QA/QC logs, with source traceability back to raw input files №1-78. Final package must include the exact raw input filename reference from Section 1A for every evidence layer/report/table/map.
Software / equipment	QGIS, Office, PDF export, archive/checksum tools.

Processing folder structure

```
99_Final_Deliverables/
├── 01_Reports
├── 02_GIS_GPKG_QGIS_QField
├── 03_Remote_Sensing_Products
├── 04_Drone_LiDAR_Orthomosaic_PointCloud
├── 05_Field_Forms_and_pXRF_Registers
├── 06_Assay_and_QAQC_Tables
├── 07_Target_Ranking_and_Decision_Matrix
├── 08_Follow_Up_Work_Plans
└── 09_Final_Report_Package
```

Step-by-step methodology

69. All reports, GIS, remote sensing, drone/LiDAR, field forms, assay/QAQC, target ranking, follow-up work plans and final report package-г зохион байгуулж өгнө.
70. Final deliverables EPSG:32647 standard CRS-тэй байна; raw/native CRS болон metadata-г хадгална.
71. Deliverable бүрт source, processing date, operator/reviewer, QA/QC status, limitation note бичнэ.

QA/QC check

QA/QC item	Acceptance note
Folder structure complete	Recorded in phase QA/QC log; reviewer/date/decision required.
Files named consistently	Recorded in phase QA/QC log; reviewer/date/decision required.
Metadata and limitations included	Recorded in phase QA/QC log; reviewer/date/decision required.
Final QA/QC notes included	Recorded in phase QA/QC log; reviewer/date/decision required.

Expected outputs

- Final report package, GIS package, remote sensing products, drone/LiDAR products, field forms, assay/QAQC tables, target ranking and follow-up plans.

Decision gate / next phase condition

- Final package is internally consistent, QA/QC reviewed and ready for management/technical review.

3. Remote sensing special subworkflows

Subworkflow	Key method	Important limitation
Sentinel-2 SNAP 13.0.0	Raw/received status check; L1C -> Sen2Cor L2A; subset boundary + 500 m/1 km; 10 m resample; cloud/shadow/vegetation mask; Natural RGB, SWIR-NIR-Red, geology composite, lithology ratio/index, NDVI, NDWI, iron oxide, ferric, clay/SWIR, ferrous, brightness index; export GeoTIFF EPSG:32647.	Sentinel output нь field validation support бөгөөд хүдэржилтийн баталгаа биш.
ASTER workflow v5	HDF import/band extraction; UTM47 project grid; b*_project band-aac alteration/lithology index тооцох; haze/edge filter зөвхөн visual; lithology composite -> favorable polygon -> raster -> normalized score; save score_porphyry_alteration_raw_v1.tif, porphyry_potential_class_v1.tif, porphyry_final_target_binary_mask_v1.tif.	Raw score=Float32 continuous; class=1/2/3; mask=0/1. Binary mask нь хүдэржилтийн баталгаа биш.
KOMPSAT-2	PAN/MS band, RPC, EPH, TXT metadata bundle; PAN/MS alignment; orthorectification; true color, false color, NDVI, pan-sharpened basemap, lineament/outcrop/access interpretation.	SWIR байхгүй тул clay/sericite/carbonate alteration-ийг дангаар батлахгүй.
ALOS-PALSAR / ASTER GDEM / DEM	Hillshade, slope, aspect, contour, drainage, watershed, ruggedness, curvature; drone flight, access, trench/drill pad, safety, lineament support.	DEM derivative нь structural support; field confirmation шаардлагатай.

4. Deposit model candidate screening table

Deposit model candidate	Supporting evidence to look for	Missing evidence / risk	Recommended validation work
Au-Cu hydrothermal vein	Quartz vein, pyrite/chalcopyrite/malachite, Au-Cu-Ag-As-Bi, shear/fault corridor, lineament intersection.	Au pXRF unreliable; vein continuity/width/grade may be unknown.	Recon mapping, rock chip/channel, lab Au fire assay + multi-element, LiDAR/structural mapping.
Intrusion-related Cu-Au-Mo	Diorite/granodiorite/gabbrodiorite contact, Cu-Mo-Bi-As, stockwork/quartz veinlets, ASTER/Sentinel alteration support.	Porphyry-style alteration zoning and sulphide system not confirmed.	ASTER validation, mapping of intrusive phases, soil grid, IP/magnetic, trench/channel.
Skarn/contact metasomatic Cu-Au-W-Bi	Intrusive-carbonate contact, garnet/epidote/magnetite/skarn minerals, W-Bi-Cu-Au association.	Carbonate/contact continuity and skarn mineral assemblage may be unclear.	Detailed contact mapping, pXRF W/Bi/Cu screening, petrography, magnetic/IP support.
Polymetallic vein	Pb-Zn-Cu-Ag-As association, vein/shear structures, gossan/iron oxide, historical occurrence overlap.	Depth continuity and grade continuity unknown.	Rock chip/channel, soil grid, structure mapping, lab Pb-Zn-Ag multi-element.
VMS-type sulphide possibility	Volcanic-sedimentary package, Cu-Zn-Pb-Ba-Fe-Mn, stratiform sulphide or gossan, regional arc/oceanic context.	Stratigraphic control and massive sulphide textures not confirmed.	Detailed stratigraphic mapping, geochemistry, IP, magnetic, targeted trenching.
Heavy mineral / placer indicator	Historical shlich/heavy mineral anomaly, drainage concentration, Au/W/Sn/Ti/Cr indicators.	Source may be transported/secondary; bedrock source not proven.	Drainage follow-up, upstream sampling, heavy mineral panning, geomorphology and bedrock checking.

5. Preliminary and final target ranking matrix

Evidence	Weight	High score criterion	Required attribute fields
Geology / lithology / intrusive contact	20	Favorable host, intrusive contact, skarn/contact, volcanic/intrusive package, mapped mineralized zone.	score_geology, confidence, data_gap, next_action
Historical geochemistry / shlich / stream sediment	15	Cu-Au-Ag-Mo-Bi-As-Pb-Zn-W-Sn anomaly overlap and drainage source consistency.	score_historical_geochem, source_scale, limitation
ASTER / Sentinel alteration and lithology	15	Clay/sericite/silica/ferric/chlorite/carbonate indicators and lithology index overlap.	score_rs, rs_product, mask_flag
Structure / lineament / intersection	15	Fault/shear/vein trend, lineament intersection, contact-parallel structures.	score_structure, trend, intersection_type
Field mapping and pXRF response	15	Malachite, pyrite, epidote, quartz vein, gossan, elevated Cu/Pb/Zn/As/Mo/W/Sn.	score_field_pxrf, instrument, qa_status
Drone LiDAR / photogrammetry evidence	8	Outcrop exposure, vein trace, trench/pit evidence, micro-topography, access confirmation.	score_drone, orthomosaic_id, lidar_id
CMCS nearest deposit / metallogenic context	7	Relevant Au-Cu/Cu-polymetallic/skarn/VMS occurrence within 5/10/20 km context.	score_cmcs, buffer_km, context_only_flag
Access / safety / workability	5	Field access possible, slope moderate, drone/sampling/trenching feasible.	score_access, hse_risk, route_status

Target class	Score range	Meaning	Action
A	>=75	Field/lab follow-up priority with multiple evidence types.	Drone/recon/sampling/trench/geophysics planning.
B	55-74	Promising but additional mapping/sampling needed.	Drone/recon + targeted sampling; update confidence.
C	35-54	Data gap or low confidence.	Limited check, additional desktop/field validation.
D	<35	Archive/monitor unless new evidence emerges.	No immediate field cost except opportunistic check.

6. Portable XRF QA/QC and register schema

Daily pXRF step	Olympus Vanta M / Bruker Titan S1 procedure	Record field
Warm-up	Instrument manufacturer recommended warm-up; battery/mode/profile check.	instrument_model, serial_no, operator, date_time
Calibration/check sample	Daily start/end check; known reference material comparison.	crm_id, expected_value, measured_value, pass_fail
Blank	Contamination check between high-grade or dusty samples.	blank_id, measured_elements, pass_fail
Duplicate/cross-check	10-15% duplicate reading or critical station repeat; Vanta M vs Titan S1 comparison.	duplicate_id, parent_sample_id, instrument_pair
Reading conditions	Surface prep, moisture, grain size, weathering, measurement window and time.	moisture, surface_condition, reading_time_sec, mode
Element suite	Cu, Pb, Zn, As, Mo, W, Sn, Mn, Fe, S and relevant pathfinders. Au not decision-grade.	element_ppm_pct fields, reliability_flag

7. Sampling methodology and QA/QC insertion

Sample type	When to use	Sample ID convention	Critical note
Representative rock chip	Alteration zone/host rock characterization.	BUD-RC-001	Avoid only high-grade visible pieces unless coded as selective.
Selective rock chip	Visible sulphide/malachite/quartz vein/mineralized float.	BUD-RC-001	Clearly flag selective bias.
Float sample	Mineralized float where outcrop is limited.	BUD-RC-001 / BUD-FLT-001 optional	Do not use as in-situ proof unless source traced.
Channel sample	Vein/zone width measurable and safe to cut across.	BUD-CH-001	Record width, orientation, recovery, continuity.
Orientation soil	Before systematic grid to test response.	BUD-SOIL-001	Depth/horizon/mesh/fraction required.
Stream sediment follow-up	Historical drainage anomaly source checking.	BUD-STR-001	Use catchment logic and upstream/downstream control.
Heavy mineral follow-up	Shlich/heavy mineral anomaly verification.	BUD-HM-001	Record panning/concentrate method and geomorphic setting.
QA/QC standard	Certified reference material insertion.	BUD-QC-STD-001	CRM suitable for expected element suite.
QA/QC blank	Contamination monitoring.	BUD-QC-BLK-001	Insert after high-grade or regular interval.
QA/QC duplicate	Field/lab precision check.	BUD-QC-DUP-001	Blind duplicate preferred.

8. Final target description sheet schema

Field	Required content
target_id	Unique ID, e.g., BUD-TGT-A01.
location	Easting/Northing EPSG:32647, license relation, access route.
evidence_summary	Concise multi-evidence summary with source layers.
geology	Host lithology, intrusive/contact relation, map confidence.
structure	Fault/shear/vein trend, intersection, LiDAR/field confirmation.
alteration	ASTER/Sentinel support, field alteration, confidence.

geochemistry	pXRF/lab/soil/stream/heavy mineral values and QA/QC status.
remote_sensing_support	Sentinel/ASTER/KOMPSAT/DEM product IDs and limitation.
drone_lidar_support	Orthomosaic tile, LiDAR DTM/DSM/lineament/outcrop evidence.
sampling_result	Rock chip/channel/soil assay summary and QA/QC performance.
confidence	High/Medium/Low with reason.
risk_data_gap	Missing evidence and uncertainty.
recommended_follow_up	Mapping, sampling, trench, IP, magnetic, scout drilling.
go_no_go_decision	Go / Conditional Go / No-Go with reviewer/date.

Appendix E — Historical Scanned Maps QGIS Vectorization Workflow v02 Detailed

Энэ appendix нь XV023222_Buduunkhad_HistoricalScannedMaps_QGIS_Vectorization_Workflow_MN_v02_Detailed.docx баримт бичгийн аргачлалыг үндсэн 78Inputs v2 Enhanced workflow-д нэмсэн хэсэг юм. Historical scan-derived vector evidence нь field/lab confirmed evidence биш бөгөөд confidence flag, data gap, scale-use limitation-тайгаар ашиглагдана.

XV-023222 / Buduunkhad / L23222

Historical Scanned Maps to Georeferenced Raster and Vector GIS Evidence Database Workflow

QGIS / GeoPackage / QA-QC / Confidence Ranking аргачлал - v02 Detailed

Энэ хувилбар нь v01 аргачлалыг илүү дэлгэрүүлж, QGIS дээр хийх бодит алхам, layer schema, Excel register sheet, QA/QC шалгуур, confidence scoring, data gap, handover acceptance criteria-г нэг бүрчлэн оруулсан audit-ready ажлын заавар юм.

Талбар	Утга
Project	XV-023222 / Buduunkhad / L23222
Workflow title	Historical Scanned Map to Georeferenced Raster and Vector GIS Evidence Database Workflow
Scope	1987-2021 historical scanned geology, geochemistry, heavy mineral, stream sediment, mineral resources, metallogenic, prospectivity and source material maps
Standard CRS	WGS 84 / UTM Zone 47N, EPSG:32647
Software	QGIS, GeoPackage, Excel register, QA/QC workbook
Workflow status	Raw Scan -> Inventory -> Georeference -> Vector Digitizing -> Register -> QA/QC -> Confidence Ranking -> Master GIS Handover
Version	v02 Detailed
Prepared date	2026-05-26

Агуулгын товч жагсаалт

1. Зорилго ба хамрах хүрээ

Энэхүү аргачлал нь Бүдүүн хад / XV-023222 / L23222 төслийн бүх historical scanned map файлыг QGIS дээр бүртгэх, georeference хийх, vector GIS layer болгон боловсруулах, Excel/GeoPackage register үүсгэх, QA/QC болон confidence ranking хийж Master GIS database-д audit-ready байдлаар нэгтгэх зорилготой.

Энэ нь ганц ашигт малтмалын зураг боловсруулах workflow биш. 1987 оны 1:200,000 шлих, ёроолын сорьц, геологи, ашигт малтмалын зураг; 2013 оны 1:50,000 геологи, ашигт малтмал, хэтийн төлөв, эх материалын зураг; 1:100,000-1:500,000 металлогени болон regional metallogenic report scan/pdf файлуудыг нэгэн зэрэг хамарна.

Raw scan-derived vector data нь historical interpretation evidence бөгөөд field/lab confirmed evidence биш. Field validation, pXRF screening, rock chip/soil sampling, laboratory assay-аар баталгаажихаас өмнө decision-grade evidence гэж ашиглахгүй.

1:50,000 зураг нь target-scale interpretation болон QField field validation-д илүү өндөр ач холбогдолтой. 1:100,000-1:500,000 зураг нь regional context, drainage/geochemical dispersion, metallogenic framework, structural trend, target screening-д ашиглагдана.

Бүх final spatial output EPSG:32647-д хадгалагдана. Native/raw CRS, source scale, GCP, georeference confidence, digitizing confidence-г metadata/register-д заавал хадгална. Raw archive файлыг засварлахгүй. Processing зөвхөн working copy дээр хийгдэнэ.

Хамрах зүйл	Энэ workflow-д хийх ажил	Хязгаарлалт
Raw scan JPG/PDF	Inventory, working copy, map type classification, legend linkage	Raw file дээр overwrite хийхгүй
Main map	Georeference, GeoTIFF, vector digitizing	Map scale-ийн хязгаарыг хадгална
Legend scan	Symbol dictionary, domain/lookup table, interpretation rule	Ихэвчлэн georeference хийхгүй
Vector evidence	Point/line/polygon layer, source traceability, QA/QC	Historical only гэсэн validation_status хадгална

Хамрах зүйл	Энэ workflow-д хийх ажил	Хязгаарлалт
Register/QAQC	Excel workbook, confidence ranking, data gap register	Хоосон field-тэй output handover хийхгүй

2. Reference document-тэй нийцүүлэх зарчим

Reference	Энэ workflow-д тусгах шаардлага	v02-д нэмсэн дэлгэрүүлэлт
Overall 78-input exploration workflow	78 raw input evidence group-ийг Phase 1 Data Audit and Master GIS Setup-тэй уялдуулах; EPSG:32647 CRS; raw data-г өөрчлөхгүй; Master GIS database; Phase 3/4/6/7/8/9 handover.	21 scan map-ыг evidence group, map family, priority, handover use-ээр ангилсан.
Phase 1 - Data Audit and Master GIS Setup	File inventory, metadata register, sidecar check, CRS check, georeference audit, GCP table, residual report, Master GeoPackage, Master QGIS project, QA/QC log, confidence ranking, data gap register.	GCP table schema, residual acceptance, raster/vector confidence score, data gap fields, acceptance criteria-г дэлгэрүүлсэн.
2013 MineralOccurrenceMap QGIS workflow	Raw scan -> GeoTIFF -> vector digitizing -> register -> QA/QC гэсэн логикийг бүх scanned map family-д өргөтгөх.	Ганц occurrence point биш: geology, structure, geochemistry anomaly, heavy mineral, source material, prospectivity, metallogenic context layer-үүдийг нэмсэн.

3. Ажил гүйцэтгэх ерөнхий sequence

94. Raw archive-г read-only гэж үзэж, бүх scan map-ыг evidence group-ээр шалгана.
95. 01_Input_Working_Copy руу файлуудыг хуулж, filename, size, extension, checksum, source note бүртгэнэ.
96. Map Inventory болон Map-to-Legend Linkage register үүсгэнэ.
97. QGIS project-ийг EPSG:32647 CRS-тэй үүсгэж, license boundary болон buffer layer-уудыг reference болгон load хийнэ.
98. Main map бүрийн georeference priority тогтоож, GCP сонголтын төлөвлөгөө гаргана.
99. QGIS Georeferencer дээр map бүрийг GeoTIFF болгож, GCP table, residual report, screenshot/check map хадгална.
100. Georeferenced raster бүрт CRS, extent, RMSE/residual, grid alignment, license/basemap/DEM overlay шалгаж raster confidence өгнө.
101. Map type бүрийн digitizing rule болон legend-based symbol dictionary-г баталгаажуулна.
102. GeoPackage layer-үүдийг үүсгэж, common source traceability fields + layer-specific fields нэмнэ.
103. Vector digitizing хийж, QGIS form tab, domain/lookup, required field constraints тохируулна.
104. Topology, geometry validity, duplicate, NULL, ID uniqueness, scale-use flag, historical/confirmed separation QA/QC шалгана.
105. Excel register workbook export хийж, confidence ranking болон data gap register бөглөнө.
106. Cross-map integration хийж, evidence давхцал/зөрүү/шийдвэрийн нөлөөллийг бүртгэнэ.
107. Master QGIS project, Master GeoPackage, QA/QC workbook, README, handover checklist багцалж дараагийн phase-д шилжүүлнэ.

Stage	Input	Main action	Output	QA gate
S1 Inventory	Raw scan files	Register + checksum + working copy	Map Inventory	All files accounted
S2 Legend linkage	Main map + legend	Symbol/domain mapping	Legend linkage register	Legend status assigned
S3 Georeference	Working raster	GCP + transformation + GeoTIFF	GeoTIFF EPSG:32647	Residual + overlay checked
S4 Vectorization	GeoTIFF + legend	Digitize by map type	GeoPackage layers	Geometry/attribute QA passed
S5 Register	Vector layers	Export and enrich attributes	Excel workbook	Required sheets complete
S6 Integration	All evidence layers	Cross-map overlay	Confidence/data gap/handover	Acceptance criteria passed

4. Input scanned map inventory

Доорх 21 файлыг нэг Map Inventory-д бүртгэж, main map, legend, report excerpt, regional context гэсэн холбоосоор удирдана. Файл бүрийн raw path, working copy path, checksum, open status, georeference status, output status-г workbook-д нэмэлт баганаар хадгална.

Map_ID	Evidence group	Raw filename	Year	Sheet	Scale	Map type	Legend	Expected output	Priority	Main use
BK-SCAN-001	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Heavy mineral sampling results map	BK-SCAN-002	GeoTIFF + heavy mineral sample/anomaly layers	P2	Шлихийн сорьц, indicator mineral, тархалтын contour
BK-SCAN-002	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_HeavyMineralSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Heavy mineral legend	BK-SCAN-001	Symbol dictionary / lookup	P2	Шлих/индикатор минералын таних тэмдэг
BK-SCAN-003	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Legend_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Stream sediment legend	BK-SCAN-004	Symbol dictionary / lookup	P2	Ёроолын сорьц, сарнилын урсгал, contour
BK-SCAN-004	04_HeavyMineral_StreamSediment_Field	1987_MN_L47-XIX_StreamSedimentSamplingResultsMap_Polyelement_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Stream sediment polyelement map	BK-SCAN-003	GeoTIFF + anomaly polygon/contour layers	P2	Cu Pb Zn Ag As Bi W Sn Mo Mn Ba F anomaly
BK-SCAN-005	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Regional geological map	BK-SCAN-006	GeoTIFF + regional geology/structure layers	P2	Региональ геологи, lithology, structure
BK-SCAN-006	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_GeologicalMap_Legend_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Regional geological legend	BK-SCAN-005	Symbol dictionary / lookup	P2	Stratigraphy, intrusion, structure, lithology
BK-SCAN-007	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Mineral resources map	BK-SCAN-008	GeoTIFF + occurrence/resource layers	P2	Региональ илрэл, гажил, хүдрийн талбай
BK-SCAN-008	05_Geology_Mineral_Prospectivity	1987_MN_L47-XIX_MineralResourcesMap_Legend_1-200000_v01_raw-scan.jpg	1987	L47-XIX	1:200,000	Mineral resources legend	BK-SCAN-007	Symbol dictionary / lookup	P2	Ашигт малтмал, элемент, илрэл,

Map_ID	Evidence group	Raw filename	Year	Sheet	Scale	Map type	Legend	Expected output	Priority	Main use
										гажлын тэмдэглэгээ
BK-SCAN-009	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Detailed geological map	BK-SCAN-010	GeoTIFF + detailed geology/structure layers	P1	Нарийвчилсан геологи, lithology, fault, section
BK-SCAN-010	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_GeologicalMap_Legend_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Detailed geological legend	BK-SCAN-009	Symbol dictionary / lookup	P1	Stratigraphy, intrusive, vein, alteration
BK-SCAN-011	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_MineralOccurrenceMap_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Mineral occurrence map	BK-SCAN-010/BK-SCAN-015	GeoTIFF + mineral occurrence/target layers	P1	Au-Cu Cu Mo As Zn илрэл + геологийн суурь
BK-SCAN-012	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessment_ReportExcerpt_B3-TolKhar_v01_raw-photo.jpg	2013	L47-74-A	Report/photo	Prospectivity report excerpt	BK-SCAN-013	Text evidence register	P1	Б-2/В-3/В-4 талбайн тайлбар
BK-SCAN-013	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_ProspectivityAssessmentMap_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Prospectivity assessment map	BK-SCAN-012	GeoTIFF + prospectivity polygons	P1	Б-3 Толь хяр, Г-1 хэтийн төлөвтэй хэсэг
BK-SCAN-014	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Source materials map	BK-SCAN-015	GeoTIFF + routes/obs/sample/trench layers	P1	Маршрут, ажиглалт, сорьц, суваг, шурф, зүсэлт
BK-SCAN-015	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-74-A_SourceMaterialsMap_Legend_1-50000_v01_raw-scan.jpg	2013	L47-74-A	1:50,000	Source materials legend	BK-SCAN-014	Symbol dictionary / lookup	P1	Ажиглалтын цэг, маршрут, сорьц, шурф, суваг
BK-SCAN-016	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MineralDistributionPatternMap_1-100000_v01_raw-scan.jpg	2013	L47-73-74	1:100,000	Mineral distribution pattern map	None	GeoTIFF + ore district/node context	P3	Металлогенийн бүс, хүдрийн дүүрэг, зангилаа
BK-SCAN-017	05_Geology_Mineral_Prospectivity	2013_MN_Namalzakh_L47-73-74_MetallogenicSchemeAndMetallogenogram_1-400000_v01_raw-scan.jpg	2013	L47-73-74	1:400,000	Metallogenic scheme/metallogenogram	None	GeoTIFF + metallogenic context	P3	Хүдрийн формац, нас, металлогенийн бүс
BK-SCAN-018	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_Legend_RawScan_2020_v01.jpg	2020	L47B Talshand	1:500,000	Regional metallogenic legend	BK-SCAN-019	Symbol dictionary / lookup	P4	Металлогений таних тэмдэг
BK-SCAN-019	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_L47B_Talshand_1M500K_RawScan_2020_v01.jpg	2020	L47B Talshand	1:500,000	Regional metallogenic map	BK-SCAN-018	GeoTIFF + regional metallogenic zones	P4	Монгол улсын 1:500k металлогени
BK-SCAN-020	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book01_ProjectBook13_1M500K_RawScan_2021_v01.pdf	2021	Regional	1:500,000	Metallogenic report book 01	BK-SCAN-019	Report evidence register	P4	Тайлангийн текст, тайлбар
BK-SCAN-021	06_Regional_Metallogenic_L47B	Regional_MetallogenicMap_Report_Book04_ProjectBook16_1M500K_RawScan_2021_v01.pdf	2021	Regional	1:500,000	Metallogenic report book 04	BK-SCAN-019	Report evidence register	P4	Тайлангийн текст, тайлбар

4.1 Inventory-д заавал нэмэх metadata баганууд

Багана	Төрөл	Заавал эсэх	Тайлбар / domain
raw_path	Text	Yes	00_Raw_Files_Archive доторх эх файл
working_copy_path	Text	Yes	01_Input_Working_Copy доторх боловсруулалтын хуулбар
file_size_mb	Decimal	Yes	Хэмжээг audit-д хадгална

Багана	Төрөл	Заавал эсэх	Тайлбар / domain
checksum_sha256	Text	Recommended	Raw vs working copy integrity шалгах
open_status	Text	Yes	Opens / Error / Needs conversion
scan_quality	Text	Yes	Good / Fair / Poor
has_coordinate_grid	Text	Yes	Yes / No / Partial / Unknown
georef_required	Text	Yes	Yes / No / Context only
georef_status	Text	Yes	Not started / In progress / Completed / Failed / Needs verification
vectorization_status	Text	Yes	Not started / Draft / Checked / Approved / Not applicable
handover_status	Text	Yes	Ready / Use with caution / Hold / Exclude

5. Map-to-legend linkage ба symbol dictionary

Main map болон legend файлыг салгаж хадгалах боловч digitizing rule, attribute domain, symbol dictionary үүсгэхэд хооронд нь заавал холбож бүртгэнэ. Legend файлыг тусад нь georeference хийх шаардлагагүй, харин map symbol тайлах evidence болгон бүртгэнэ.

Main map	Legend / report support	Digitizing rule	Domain / lookup үүсгэх зүйл
1987 HeavyMineralSamplingResultsMap	1987 HeavyMineralSamplingResultsMap_Legend	Шлих, indicator mineral, contour, sample symbol domain үүсгэнэ.	mineral_indicator, anomaly_class, sample_symbol
1987 StreamSedimentSamplingResultsMap_Polyelement	1987 StreamSedimentSamplingResultsMap_Legend	Element suite, anomaly contour, drainage dispersion symbol domain үүсгэнэ.	element_suite, anomaly_level, contour_type
1987 GeologicalMap	1987 GeologicalMap_Legend	Geological unit, age, lithology, intrusive, structure symbol domain үүсгэнэ.	map_symbol, lithology, age, intrusive_type
1987 MineralResourcesMap	1987 MineralResourcesMap_Legend	Commodity, occurrence type, anomaly, ore field symbol domain үүсгэнэ.	commodity, occurrence_type, ore_field_type
2013 GeologicalMap	2013 GeologicalMap_Legend	Detailed lithology, fault, vein, alteration, section symbol domain үүсгэнэ.	stratigraphic_unit, vein_type, alteration
2013 MineralOccurrenceMap	2013 GeologicalMap/SourceMaterials legend болон map label	Au-Cu, Cu, Mo, As, Zn occurrence point and target zone digitizing rule үүсгэнэ.	commodity_group, target_style
2013 SourceMaterialsMap	2013 SourceMaterialsMap_Legend	Route, station, sample, trench/pit/shaft/channel/section symbol domain үүсгэнэ.	observation_type, sample_type, work_type
Regional MetallogenicMap L47B	Regional MetallogenicMap_L47B_Legend + Book01/Book04	Metallogenic belt, ore district, commodity group, regional occurrence context rule үүсгэнэ.	metallogenic_unit, ore_formation, scale_context

5.1 Symbol dictionary үүсгэх алхам

108. Legend scan-ыг QGIS эсвэл image viewer дээр нээгээд symbol бүрийг screenshot/zoom түвшинд шалгана.

109. Symbol_Code, Symbol_Image_Ref, Mongolian_Name, English_Name, Geometry_Type, Default_Layer, Attribute_Field, Domain_Value, Notes гэсэн баганатай dictionary sheet үүсгэнэ.
110. Main map дээрх тэмдэглэгээ legend-тэй тохирч байгаа эсэхийг 10-20 жишээ feature дээр туршиж баталгаажуулна.
111. Тодорхойгүй тэмдэглэгээ бүрт attribute_confidence = Low/Unknown, data_gap_type = Unclear legend гэж тэмдэглэнэ.
112. QGIS layer form дээр dropdown domain болгон ашиглах lookup list үүсгэнэ.

Symbol dictionary багана	Жишээ	Тайлбар
symbol_code	HM-MAG-01	Дотоод код
symbol_name_mn	Магнетит агуулсан шлих	Legend-ээс уншсан нэр
symbol_name_en	Magnetite heavy mineral indicator	Тайлан/олон улсын нэршил
geometry_type	Point / Line / Polygon	Digitize хийх геометр
target_layer	heavy_mineral_sample_points	QGIS layer
domain_field	mineral_indicator	Атрибут талбар
domain_value	Magnetite	Нэг мөр стандарт утга
confidence_default	Medium	Тэмдэглэгээний тод байдал
notes	Legend scan unclear	Тайлбар

6. Folder structure ба file governance

Дараах бүтэц нь Phase 1 Data Audit and Master GIS Setup дотор ажиллахад тохиромжтой. Raw archive-г өөрчлөхгүй; энэ бүтэц нь working copy болон output-д зориулагдана.

```
01_Phase_1_Data_Audit_and_Master_GIS_Setup/
  00_Admin_and_Method/
  01_Input_Working_Copy/
    01_HeavyMineral_StreamSediment_1987/
    02_Geology_MineralResources_1987/
    03_Geology_MineralOccurrence_Prospectivity_2013/
    04_Metallogenic_Regional_2013_2021/
  02_Inventory_and_Metadata/
  03_CRS_Check/
  04_Georeference_Check/
    01_GCP_Tables/
    02_Georeferenced_Rasters/
    03_Residual_Reports/
    04_Georeference_Screenshots/
    05_Low_Confidence_Georef/
  05_Vector_Digitized/
    01_Geology_Polygons/
    02_Structures_Faults_Lines/
    03_Mineral_Occurrences_Points/
    04_Geochemistry_Anomaly_Polygons/
    05_HeavyMineral_StreamSediment_Layers/
    06_Source_Materials_Points_Lines/
```

07_Prospectivity_Target_Zones/
08_Metallogenic_Context/
06_Register_Metadata/
07_QGIS_Project/
08_QAQC_and_Confidence/
09_Handover_Package/

Governance rule	Зорилго	Хэрэгжүүлэх арга
Raw read-only	Эх өгөгдөл эвдрэхээс хамгаалах	Raw archive дээр бичих эрхгүй, processing зөвхөн copy дээр
Checksum	Файл солигдсон эсэхийг хянах	SHA-256 raw ба working copy-д хадгалах
Versioning	Дахин боловсруулалтыг ялгах	v01, v02, v03; draft файлыг _DRAFT гэж тэмдэглэх
Sidecar grouping	Raster metadata алдагдахаас сэргийлэх	.aux.xml, .ovr, .tfw, .jgw, .rpc, .eph файлыг хамт хадгалах
Change log	Audit trail үүсгэх	Action_Log.xlsx: date/operator/action/input/output/status

7. QGIS project setup

113. QGIS -> Project -> New сонгоно.
114. Project -> Properties -> CRS хэсгээс WGS 84 / UTM Zone 47N, EPSG:32647 сонгоно.
115. Distance unit = meters; Area unit = square kilometers/hectares; Ellipsoid = WGS84 тохируулна.
116. Project-г 07_QGIS_Project хавтаст XV023222_Buduunkhad_HistoricalScannedMaps_Vectorization_QGIS_EPSG32647_v02.qgz нэрээр хадгална.
117. GeoPackage connection үүсгэж Master GeoPackage-г Browser Panel-д холбоно.
118. Layer group structure-г доорх байдлаар үүсгэнэ.
119. Project Properties -> General дээр project title, author/team, path relative тохиргоог шалгана.
120. Project Properties -> Data Sources дээр Automatically create transaction groups шаардлагатай бол идэвхжүүлнэ.
121. QGIS snapping болон topology editing тохиргоог vector digitizing эхлэхээс өмнө тохируулна.

Layer group structure:
01_Source_Raw_and_Georef_Raster
02_Admin_Boundary_and_Buffer
03_Geology
04_Mineral_Occurrence
05_HeavyMineral_StreamSediment
06_Source_Materials
07_Prospectivity_Targets
08_Metallogenic_Context
09_QAQC_Confidence
10_Handover

QGIS setting	Recommended value	Тайлбар
Project CRS	EPSG:32647	Final output CRS
On-the-fly reprojection	Enabled	Native layer display хийхэд
Snapping type	Vertex and segment	Line/polygon digitizing-д
Snapping tolerance	5-10 pixels / scale dependent	Scale-аас хамааруулж тохируулна
Topological editing	Enabled for line/polygon	Overlap/sliver багасгана
Avoid overlap	Enabled for same polygon layers	Geology/target zone polygon-д
Default encoding	UTF-8	Монгол/кирилл attribute алдагдахгүй
Project paths	Relative	Handover package зөөврлөхөд

8. Georeferencing workflow

Legend scan файлыг гол төлөв georeference хийхгүй. Main map scan-уудыг georeference хийж GeoTIFF болгосны дараа legend scan-ыг symbol dictionary, lookup/domain, attribute coding хийхэд ашиглана.

8.1 GCP сонгох эх сурвалжийн эрэмбэ

Эрэмбэ	Control source	Ашиглах нөхцөл	Confidence impact
1	Coordinate grid intersection	Зураг дээр coordinate grid тод, огтлолцол сайн харагдаж байвал хамгийн түрүүнд сонгоно.	High
2	Map frame corner coordinate / labelled tick	Булан/захын coordinate тодорхой үед.	High
3	Map sheet boundary	L47-XIX, L47-74-A sheet boundary coordinates баталгаатай үед.	Medium-High
4	License boundary corner / known control point	Тусгай зөвшөөрлийн хил эсвэл баталгаатай vector boundary давхцаж байвал.	Medium
5	Stable topographic feature	Голын уулзвар, замын огтлолцол, ridge/valley гэх мэт.	Medium-Low
6	Visual matching only	Grid/coordinate байхгүй үед зөвхөн context map-д.	Low / Needs verification

8.2 QGIS Georeferencer дээр хийх алхам

122. Raster -> Georeferencer нээнэ.
123. Open Raster сонгож working copy JPG/PDF raster-г нээнэ. PDF бол шаардлагатай бол өмнө нь 300-600 dpi TIFF/PNG болгон хөрвүүлнэ.
124. Settings -> Transformation Settings нээгээд Target CRS = EPSG:32647 сонгоно.
125. Output raster нэрийг 04_Georeference_Check/02_Georeferenced_Rasters хавтаст naming standard-аар өгнө.
126. Load in QGIS when done сонголтыг идэвхжүүлнэ.
127. Compression = LZW; creation options-д TILED=YES, COMPRESS=LZW гэж боломжтой бол өгнө.
128. GCP цэгүүдийг бүх талбайд жигд тархааж оруулна. Зөвхөн 4 булангаар хязгаарлахгүй.

129. Coordinate оруулахдаа эх coordinate нь longitude/latitude бол decimal degree болгож оруулаад QGIS transform ашиглана эсвэл UTM координатаар шууд оруулна.
130. Transformation-г эхлээд Polynomial 1 ашиглаж туршина. Residual их, scan distortion local байвал Thin Plate Spline туршиж, confidence note бичнэ.
131. Georeferencing ажиллуулсны дараа GeoTIFF layer properties -> Source дээр CRS, extent, pixel size, path шалгана.
132. GeoTIFF-г license boundary, DEM hillshade, basemap, өмнө georeferenced raster-тай давхарлаж overlay QA хийнэ.
133. GCP table-г CSV/XLSX болгон хадгалж, residual report screenshot болон QGIS map screenshot хадгална.

Scale	Minimum / preferred GCP	Transformation	Use rule	Confidence анхааруулга
1:50,000	Minimum 6; preferred 8-12	Polynomial 1 / Helmert / TPS only if needed	Target-scale digitizing-д ашиглана	GCP муу бол occurrence/target vector confidence буурна
1:100,000	6-10	Polynomial 1	Ore district / mineral distribution context	Local target geometry гэж хэтрүүлж ашиглахгүй
1:200,000	6-10	Polynomial 1	Regional geology/geochemistry/drainage evidence	Field validation before decision-grade use
1:400,000-1:500,000	4-8	Polynomial 1	Metallogenic context only	Context evidence; local target boundary биш

8.3 DMS coordinate-г decimal degree болгох дүрэм

Зарим scan map дээр coordinate нь 96°30′E, 46°00′N гэх мэт DMS хэлбэртэй байдаг. Decimal degree = degree + minute/60 + second/3600. East/North эерэг, West/South сөрөг тэмдэгтэй байна.

DMS	Decimal degree	Тайлбар
96°30′00″E	96.500000	96 + 30/60
45°50′00″N	45.833333	45 + 50/60
96°45′00″E	96.750000	96 + 45/60
46°00′00″N	46.000000	46 + 0/60

9. Raster QA/QC ба confidence

QA/QC item	QGIS дээр шалгах арга	Pass criteria	Fail / action
CRS	Layer Properties -> Source	EPSG:32647	Wrong CRS бол reproject/georef дахин
Extent	Layer Properties -> Information + map canvas	License/buffer/map sheet-тэй logical overlap	Outside extent бол GCP/check coordinate
GCP count	Georeferencer GCP table	Scale-specific minimum хангагдсан	GCP нэмэх
Residual/RMSE	GCP residual table	Outlier багатай, grid alignment зөв	Outlier GCP-г шалгах/хасах

Grid alignment	Coordinate grid/tick давхцал	Булан болон дунд grid сайн таарсан	Transformation/GCP дахин
Visual distortion	Raster rotation/stretch	Уншигдахуйц, суналт хэтрээгүй	TPS/scan crop дахин
Raster completeness	Canvas display / overview	Зураг тасалдаагүй, эргээгүй	Export/render setting шалгах
Metadata	Inventory + properties	Source file, scale, year, map sheet бүртгэгдсэн	Metadata register бөглөх

Raster confidence	Шалгуур	Use status
High	8+ well-distributed GCP; grid/coordinate сайн; residual acceptable; license/topographic overlay сайн; scan quality good	Target-scale digitizing болон Phase 4 ranking-д ашиглаж болно
Medium	4-8 GCP; minor distortion; overlay logical; зарим grid/scan issue байна	Overlay/interpretation-д болгоомжтой ашиглана
Low	GCP цөөн; grid тодорхойгүй; regional scale; visual match давамгайлсан	Context evidence; local target boundary болохгүй
Needs verification	CRS/georef эргэлзээтэй; residual өндөр; feature location conflict	Vector digitizing хийхээс өмнө дахин шалгана

9.1 Georeferenced raster output naming

Raster theme	Output filename
1987 Heavy mineral	XV023222_Buduunkhad_1987_L47-XIX_HeavyMineralSamplingResultsMap_1-200K_Georeferenced_EPSG32647_v02.tif
1987 Stream sediment	XV023222_Buduunkhad_1987_L47-XIX_StreamSedimentPolyelementMap_1-200K_Georeferenced_EPSG32647_v02.tif
1987 Geology	XV023222_Buduunkhad_1987_L47-XIX_GeologicalMap_1-200K_Georeferenced_EPSG32647_v02.tif
1987 Mineral resources	XV023222_Buduunkhad_1987_L47-XIX_MineralResourcesMap_1-200K_Georeferenced_EPSG32647_v02.tif
2013 Detailed geology	XV023222_Buduunkhad_2013_L47-74-A_GeologicalMap_1-50K_Georeferenced_EPSG32647_v02.tif
2013 Mineral occurrence	XV023222_Buduunkhad_2013_L47-74-A_MineralOccurrenceMap_1-50K_Georeferenced_EPSG32647_v02.tif
2013 Prospectivity	XV023222_Buduunkhad_2013_L47-74-A_ProspectivityAssessmentMap_1-50K_Georeferenced_EPSG32647_v02.tif
2013 Source materials	XV023222_Buduunkhad_2013_L47-74-A_SourceMaterialsMap_1-50K_Georeferenced_EPSG32647_v02.tif
2013 Mineral distribution	XV023222_Buduunkhad_2013_L47-73-74_MineralDistributionPatternMap_1-100K_Georeferenced_EPSG32647_v02.tif
2013 Metallogenic scheme	XV023222_Buduunkhad_2013_L47-73-74_MetallogenicSchemeMetallogenogram_1-400K_Georeferenced_EPSG32647_v02.tif
2020 Regional metallogenic	XV023222_Buduunkhad_2020_L47B_Talshand_RegionalMetallogenicMap_1-500K_Georeferenced_EPSG32647_v02.tif

10. Vectorization strategy by map type

Map type	Vector output	Use limit	Primary QA/QC
Geological maps	geology_units_50k_polygons, geology_units_200k_polygons, structures_faults_lines, intrusive_contacts_lines,	1:50k = local evidence; 1:200k = regional evidence	Polygon topology, boundary match, symbol domain

Map type	Vector output	Use limit	Primary QA/QC
	dyke_vein_lines, alteration_zones_polygons		
Mineral occurrence / mineral resources maps	mineral_occurrences_points, mineralized_zones_polygons, ore_field_prospect_polygons, occurrence_labels, commodity groups	Historical map-derived evidence; field/lab confirmed гэж нэрлэхгүй	Point placement, commodity spelling, duplicate/cross-ref
Heavy mineral sampling maps	heavy_mineral_sample_points, heavy_mineral_anomaly_polygons, indicator_mineral_distribution_polygons, drainage_interpretation_lines	Drainage/source direction interpretation requires DEM/drainage check	Contour closure, drainage relation, indicator mineral domain
Stream sediment polyelement maps	stream_sediment_sample_points, stream_sediment_anomaly_polygons, geochemical_anomaly_contours_lines, drainage_anomaly_trend_lines	Multi-element suite field заавал бөглөнө	Element suite, anomaly class, drainage consistency
Prospectivity assessment maps	prospectivity_target_zones_polygons, priority_areas, named_prospects, recommended_followup_zones	Prospect polygon = historical interpretation; ranking-д confidence-тэй хэрэглэнэ	Target ID, evidence basis, priority consistency
Source materials maps	source_material_observation_points, source_material_route_lines, sample_points, trench_pit_shaft_channel_points, section_lines	QField validation, field route planning-д шууд хэрэгтэй	Route connectivity, station/sample ID, source symbol
Metallogenic maps	metallogenic_zones_polygons, ore_district_node_polygons, regional_structure_lines, regional_occurrence_context_points	Context only; local target boundary биш	Scale flag, use_limit, regional label

11. Master GeoPackage design

Нэг Master GeoPackage үүсгэнэ: XV023222_Buduunkhad_HistoricalScannedMaps_Vectorized_MasterGIS_EPSG32647_v02.gpkg. Layer бүрийн CRS EPSG:32647 байна. Layer name lowercase_snake_case байна.

No	Layer name	Geometry	Purpose
01	scan_map_index_polygons	Polygon	Georeferenced map extent, map ID, raster confidence
02	georeference_gcp_points	Point	GCP coordinate, residual, control source
03	geology_units_50k_polygons	Polygon	2013 detailed geology units
04	geology_units_200k_polygons	Polygon	1987 regional geology units
05	structures_faults_lines	LineString	Faults, inferred faults, lineaments
06	intrusive_contacts_lines	LineString	Intrusive/contact boundaries
07	dyke_vein_lines	LineString	Dyke, vein, quartz vein, section lines
08	alteration_zones_polygons	Polygon	Alteration zones if shown
09	mineral_occurrences_points	Point	Mineral occurrence/resource points
10	mineralized_zones_polygons	Polygon	Mineralized zones / ore fields
11	ore_field_prospect_polygons	Polygon	Ore field/prospect areas
12	heavy_mineral_sample_points	Point	Heavy mineral sample points if shown
13	heavy_mineral_anomaly_polygons	Polygon	Indicator mineral/anomaly polygons

No	Layer name	Geometry	Purpose
14	stream_sediment_sample_points	Point	Stream sediment sample points if shown
15	stream_sediment_anomaly_polygons	Polygon	Polyelement anomaly polygons
16	geochemical_anomaly_contours_lines	LineString	Anomaly contour/dispersion lines
17	prospectivity_target_zones_polygons	Polygon	Prospectivity / priority target zones
18	source_material_observation_points	Point	Observation stations
19	source_material_route_lines	LineString	Field route lines
20	source_material_trench_pit_points	Point	Trench/pit/shaft/channel points
21	metallogenic_zones_polygons	Polygon	Regional metallogenic belt/zone polygons
22	regional_occurrence_context_points	Point	Regional occurrence points if shown
23	source_cross_reference	No geometry / table	Feature relationships across maps
24	data_confidence_ranking_spatial	Point/Polygon/Table	Spatial confidence features
25	data_gap_register_spatial	Point/Polygon/Table	Spatial data gaps

11.1 GeoPackage үүсгэх QGIS алхам

134. Browser Panel -> GeoPackage -> Create Database эсвэл Layer -> Create Layer -> New GeoPackage Layer сонгоно.
135. Database path: 05_Vector_Digitized/XV023222_Buduunkhad_HistoricalScannedMaps_Vectorized_MasterGIS_EPSG32647_v02.gpkg гэж өгнө.
136. Эхний layer-ийг үүсгэсний дараа дараагийн layer-үүдийг ижил database-д Add layer to existing database байдлаар нэмнэ.
137. Geometry type, CRS EPSG:32647, attribute fields-ийг schema-ийн дагуу үүсгэнэ.
138. Layer бүрт fid/geometry auto field-г user form дээр нууж, feature_id, source_map_id, confidence fields-ийг required болгоно.
139. Layer style QML файлыг 05_Master_GIS_Database/Styles_QML хавтаст хадгалж version-той нэрлэнэ.

12. Field schema ба domain/lookup

12.1 Common source traceability fields

Field	Type	Required	Purpose / domain
feature_id	Text	Yes	Unique feature ID, e.g. BUD-MIN-0001
source_map_id	Text	Yes	BK-SCAN-xxx
source_file	Text	Yes	Raw filename
source_year	Integer	Yes	1987/2013/2020/2021
map_sheet	Text	Yes	L47-XIX / L47-74-A / L47B
map_scale	Text	Yes	1:50,000 etc.
map_type	Text	Yes	Geology / mineral / heavy mineral / stream sediment
digitized_from_raster	Text	Yes	Georeferenced GeoTIFF filename
legend_file	Text	Recommended	Related legend file
original_symbol	Text	Recommended	Symbol as shown on map
original_label	Text	Recommended	Label/number on map
digitized_by	Text	Yes	Operator name
digitized_date	Date	Yes	Date of digitizing
geometry_source	Text	Yes	Domain: digitized/interpreted/approximate

Field	Type	Required	Purpose / domain
geom_confidence	Text	Yes	High/Medium/Low/Needs verification
attribute_confidence	Text	Yes	High/Medium/Low/Unknown
overall_confidence	Text	Yes	High/Medium/Low/Needs verification
qaqc_status	Text	Yes	Draft/Checked/Approved/Rejected
validation_status	Text	Yes	Historical only/Field checked/Sampled/Lab confirmed/Not found/Not applicable
recommended_followup	Text	Recommended	Field check/Rock chip/Soil grid/No action/Data gap
comment	Text	No	Free text

12.2 Standard domain values

Field	Allowed values	Тайлбар
geom_confidence	High; Medium; Low; Needs verification	Geometry/source position confidence
attribute_confidence	High; Medium; Low; Unknown	Legend/label/attribute confidence
overall_confidence	High; Medium; Low; Needs verification	Combined confidence
qaqc_status	Draft; Checked; Approved; Rejected	QA/QC workflow status
validation_status	Historical only; Field checked; Sampled; Lab confirmed; Not found; Not applicable	Field/lab verification status
geometry_source	Digitized from georeferenced historical scan; Derived from map symbol; Interpreted from contour; Approximate from regional map	Source geometry origin
scale_context	Local target-scale; Regional evidence; Metallogenic context; Report context	Map scale use limitation
use_limit	Can support target ranking; Use with caution; Context only; Do not use until verified	Use status

13. Layer-specific schema

13.x Layer: geology_units_polygons

Field name	Type	Required
unit_id	Text	Yes
map_symbol	Text	Yes
lithology	Text	Yes
age	Text	Recommended
stratigraphic_unit	Text	Recommended
intrusive_type	Text	Optional
alteration	Text	Optional
description	Text	Optional
source_scale	Text	Yes

Field name	Type	Required
confidence	Text	Yes
comment	Text	No

13.x Layer: structures_faults_lines

Field name	Type	Required
struct_id	Text	Yes
struct_type	Text	Yes
certainty	Text	Yes
trend	Text	Recommended
relation_to_mineralization	Text	Recommended
relation_to_intrusion	Text	Recommended
relation_to_geochemistry	Text	Recommended
source_symbol	Text	Recommended
confidence	Text	Yes
comment	Text	No

13.x Layer: mineral_occurrences_points

Field name	Type	Required
occ_id	Text	Yes
map_no	Text	Recommended
occ_name	Text	Optional
commodity_raw	Text	Yes
commodity_1	Text	Recommended
commodity_2	Text	Optional
commodity_group	Text	Recommended
occurrence_type	Text	Recommended
mineralization	Text	Optional
lithology	Text	Optional
structure_relation	Text	Optional
target_style	Text	Optional
recommended_followup	Text	Recommended
confidence	Text	Yes
comment	Text	No

13.x Layer: heavy_mineral_anomaly_polygons

Field name	Type	Required
anomaly_id	Text	Yes
mineral_indicator	Text	Yes
anomaly_class	Text	Recommended
source_material	Text	Recommended
drainage_relation	Text	Recommended
interpreted_source_direction	Text	Optional
confidence	Text	Yes
recommended_followup	Text	Recommended
comment	Text	No

13.x Layer: stream_sediment_anomaly_polygons

Field name	Type	Required
anomaly_id	Text	Yes
element_suite	Text	Yes
dominant_element	Text	Recommended
associated_elements	Text	Recommended
anomaly_level	Text	Recommended
drainage_basin	Text	Optional
possible_source_area	Text	Optional
confidence	Text	Yes
recommended_followup	Text	Recommended
comment	Text	No

13.x Layer: prospectivity_target_zones_polygons

Field name	Type	Required
target_id	Text	Yes
target_name	Text	Recommended
prospect_class	Text	Recommended
evidence_basis	Text	Yes
dominant_commodity	Text	Recommended
associated_commodities	Text	Optional
geology_control	Text	Recommended
structure_control	Text	Recommended
geochem_support	Text	Recommended
historical_work_support	Text	Recommended
priority	Text	Yes
data_gap	Text	Optional
recommended_next_work	Text	Recommended
confidence	Text	Yes
comment	Text	No

13.x Layer: source_material_observation_points

Field name	Type	Required
obs_id	Text	Yes
route_id	Text	Recommended
station_no	Text	Recommended
observation_type	Text	Yes
lithology	Text	Optional
mineralization	Text	Optional
sample_reference	Text	Optional
trench_pit_reference	Text	Optional
confidence	Text	Yes
comment	Text	No

13.x Layer: source_material_route_lines

Field name	Type	Required
route_id	Text	Yes
route_no	Text	Recommended
source_year	Integer	Yes
observer	Text	Optional
route_type	Text	Recommended
related_observations	Text	Optional
confidence	Text	Yes
comment	Text	No

13.x Layer: metallogenic_zones_polygons

Field name	Type	Required
zone_id	Text	Yes
zone_name	Text	Recommended
metallogenic_unit	Text	Recommended
ore_formation	Text	Recommended
commodity_group	Text	Recommended
scale_context	Text	Yes
relation_to_license	Text	Recommended
confidence	Text	Yes
use_limit	Text	Yes
comment	Text	No

14. QGIS digitizing SOP

140. GeoTIFF raster layer-ийг 40-60% opacity-тэй болгож, contrast/stretch-ийг уншигдах хэмжээнд тохируулна. Raster-г засварлахгүй.
141. Legend dictionary sheet-ийг нээлттэй байлгаж, symbol бүрийг стандарт domain value-тай тулгана.
142. Зөв layer сонгосон эсэхээ шалгаж Toggle Editing асаана.
143. Feature digitize хийхдээ map symbol-ийн төв, line-ийн гол, polygon boundary-ийн зураг дээрх бодит хүрээг баримтална.
144. Feature бүрт feature_id болон source_map_id-г шууд өгнө. Feature ID давхцахгүй байх ёстой.
145. original_symbol, original_label талбаруудыг map дээрх байдлаар бичнэ. Тодорхойгүй бол Unknown гэж бичээд data gap үүсгэнэ.
146. Confidence-г геометр болон attribute тус бүрээр өгнө. Ерөнхий confidence нь хамгийн сул confidence-оос өндөр байж болохгүй.
147. Editing дуусах бүрт Save Layer Edits хийнэ. Өдөр бүрийн төгсгөлд GeoPackage backup үүсгэнэ.
148. QA/QC reviewer шалгасны дараа qaqc_status = Checked эсвэл Approved болгож өөрчилнө.

Scale flag	Digitizing нарийвчлалын зарчим	Prohibited use
1:50,000	Local / target-scale evidence. Илрэл, маршрут, target polygon, geology contact-д ашиглаж болно.	Field/lab баталгаагүй байхад reserve/resource conclusion хийхгүй.
1:100,000	Ore district / mineral distribution context. Polygon boundary-г ойролцоогоор авна.	Local target boundary мэт ашиглахгүй.

Scale flag	Digitizing нарийвчлалын зарчим	Prohibited use
1:200,000	Regional geology/geochemistry/drainage evidence. Anomaly/structure-г regional support гэж хэрэглэнэ.	Detailed trench/soil grid location-ийг дангаар тогтоохгүй.
1:400,000-1:500,000	Metallogenic context only. Deposit model, regional belt, report figure-д хэрэглэнэ.	License доторх local target geometry гэж ашиглахгүй.

15. Layer бүрийн нарийвчилсан SOP

15.1 Geological unit polygons

2013 1:50k болон 1987 1:200k geological map-аас lithology/stratigraphy нэгжүүдийг digitize хийнэ. Эхлээд major unit boundary, дараа нь intrusive body, alteration zone, dyke/vein separate layer болгон оруулна. Polygon closure, overlap, sliver шалгана. 1:50k layer-ийг geology_units_50k_polygons, 1:200k layer-ийг geology_units_200k_polygons гэж тусгаарлана.

15.2 Structures/faults/lineaments

Fault, inferred fault, contact, lineament, shear, fold axis, vein trend зэрэг line feature-ийг structures_faults_lines layer-д оруулна. certainty = Observed/Inferred/Interpreted; trend = NE-SW/NW-SE/N-S/E-W гэх мэт. Илрэл, геохими, intrusive contact-тай хамаарал байвал relation_to_mineralization талбарт тэмдэглэнэ.

15.3 Mineral occurrence points

Mineral occurrence symbol-ийн төвд point тавина. commodity_raw-д map дээрх тэмдэглэгээг яг бичнэ; commodity_1/2-д standardized element оруулна. Au-Cu, Cu, Mo, As, Zn зэрэг spelling-ийг lookup-тай тулгана. Map no, label тодорхойгүй бол attribute_confidence = Low/Unknown.

15.4 Heavy mineral layers

Individual sample point харагдаж байвал sample point digitize хийнэ. Anomaly/indicator mineral contour харагдаж байвал polygon эсвэл line contour болгон оруулна. DEM/drainage layer-тай давхарлаж interpreted_source_direction-г зөвхөн тайлбарлагдсан үед бөглөнө.

15.5 Stream sediment polyelement layers

Polyelement anomaly contour, drainage dispersion, sample point-уудыг тусад нь digitize хийнэ. element_suite field-д Cu Pb Zn Ag As Bi W Sn Mo Mn Ba F зэрэг багцуудыг нэг мөр стандарт бичнэ. Anomaly level тодорхойгүй бол data gap үүсгэнэ.

15.6 Source materials layers

Route lines, observation station, sample point, trench/pit/shaft/channel, section line-уудыг тусдаа layer-д оруулна. Field route planning болон QField validation-д хамгийн ашигтай тул route_id, station_no, sample_reference талбарыг аль болох бүрэн бөглөнө.

15.7 Prospectivity target zones

Prospectivity map дээрх Б-3 Толь хяр, Г-1 зэрэг target/prospect zone-уудыг polygon хэлбэрээр оруулна. evidence_basis-д occurrence, geology, geochem, historical work support-ыг бичнэ. priority = P1/P2/P3/P4 гэж өгч, regional map-аас авсан polygon бол use_limit = Context only гэж тэмдэглэнэ.

15.8 Metallogenic context

1:400k-1:500k regional metallogenic map-аас belt, ore district, node, ore formation polygon/point-ийг context layer болгон оруулна. Энэ layer-ийг Phase 3 concept model болон Phase 4 scoring support-д ашиглах боловч local field target boundary биш.

16. Excel register workbook

Нэг QA/QC workbook гаргана: XV023222_Buduunkhad_HistoricalScannedMaps_Vectorization_Register_QAQC_v02.xlsx. Workbook нь GIS layer бүрийн attribute export, GCP, QA/QC, confidence, data gap, handover checklist-ийг нэг дор хадгална.

Sheet	Purpose
00_README	Workbook purpose, version, owner, CRS, definitions
01_Map_Inventory	21 scan map/report file metadata
02_Map_Legend_Linkage	Main map + legend + symbol dictionary status
03_Georeference_GCP_Table	GCP coordinates, source, residual
04_Georeference_QAQC	Raster QA/QC results and raster confidence
05_Geology_Units_Register	Geology polygon export
06_Structures_Register	Fault/structure/line export
07_Mineral_Occurrences_Register	Occurrence point export
08_HeavyMineral_Register	Heavy mineral point/polygon export
09_StreamSediment_Register	Stream sediment anomaly/sample export
10_Prospectivity_Target_Register	Prospect/target zone export
11_Source_Materials_Register	Route/obs/sample/trench export
12_Metallogenic_Context_Register	Regional metallogenic context export
13_Topology_QAQC	Geometry/topology check log
14_Attribute_QAQC	NULL/domain/ID/spelling check log
15_Confidence_Ranking	Raster/vector confidence scoring
16_Data_Gap_Register	Gap type, impact, action, owner
17_Change_Log	Date/operator/action/input/output/status
18_Handover_Checklist	Final acceptance and phase handover status
19_Source_Cross_Reference	Cross-map evidence relationship table
20_Lookups_Domains	Domain values for QGIS forms

16.1 GCP table sheet schema

Column	Type	Description
gcp_id	Text	Map_ID + sequence, e.g. BK-SCAN-009-GCP-001
source_map_id	Text	BK-SCAN-xxx
pixel_x	Decimal	Georeferencer pixel x

Column	Type	Description
pixel_y	Decimal	Georeferencer pixel y
map_x	Decimal	Target CRS Easting or longitude
map_y	Decimal	Target CRS Northing or latitude
coord_source	Text	Grid intersection / tick / boundary / topographic feature
residual_x	Decimal	Residual x
residual_y	Decimal	Residual y
residual_total	Decimal	Total residual
used_in_transform	Text	Yes / No
review_note	Text	Outlier or accepted note

17. QA/QC checklist

QA/QC group	Check item	Pass criteria
File inventory	Бүх raw file бүртгэгдсэн эсэх	21 scanned map/report file Map Inventory-д орсон
File inventory	Main map ба legend холбоотой эсэх	Map-to-legend linkage register бөглөгдсөн
File inventory	Raw file өөрчлөгдөөгүй эсэх	Raw archive checksum / working copy path бүртгэгдсэн
CRS / georeference	CRS EPSG:32647 эсэх	Final GeoTIFF/vector layer EPSG:32647
CRS / georeference	GCP хангалттай эсэх	Scale-specific GCP minimum хангагдсан
CRS / georeference	Residual error acceptable эсэх	GCP residual report хадгалагдсан, outlier note бичигдсэн
CRS / georeference	Overlay check	License boundary / basemap / DEM / other map-тэй logical overlap
Geometry	Invalid / empty geometry	QGIS Check validity OK
Geometry	Duplicate feature	Duplicate geometry / duplicate ID шалгасан
Geometry	Polygon overlap/sliver	Topology check хийсэн
Geometry	Point outside map extent/license/buffer	Flag or data gap created
Geometry	Line dangle/overshoot/undershoot	Structure/route/contact line-д reviewer check хийсэн
Attribute	Required NULL	Required fields бүрэн бөглөгдсөн
Attribute	ID uniqueness	feature_id, occ_id, anomaly_id, target_id unique
Attribute	Commodity/element spelling	Domain list-тэй нийцсэн
Attribute	Source traceability	source_map_id/source_file/digitized_from_raster бүрэн
Interpretation	Scale-use flag	Regional map-derived feature local target мэт хэтрүүлээгүй
Interpretation	Historical vs confirmed	Historical map-derived data field/lab confirmed data-тай холигдоогүй
Handover	Workbook complete	Required sheets бөглөгдсөн
Handover	README and change log	Handover note complete

18. Confidence ranking logic

Бүх raster болон vector output-д source quality, scan quality, legend clarity, GCP quality, georeference accuracy, map scale suitability, digitizing clarity, attribute completeness, cross-map consistency, field validation status гэсэн шалгуураар confidence үнэлгээ өгнө. Ерөнхий confidence нь хамгийн сул critical criterion-оос өндөр байж болохгүй.

Criterion	Score 0	Score 1	Score 2	Weight
Source quality	Unknown source	Known file but partial metadata	Known source + year + scale + map sheet	10%
Scan quality	Unreadable/poor	Usable with unclear areas	Clear symbols/labels	10%
Legend clarity	Missing	Partial/unclear	Clear linked legend	10%
GCP quality	Insufficient/weak	Minimum met	Well-distributed preferred count	15%
Georeference accuracy	High residual/conflict	Acceptable regional	Good overlay/grid	15%
Map scale suitability	Too regional	Regional/context	Target-scale/local	10%
Digitizing clarity	Approximate	Mostly clear	Clear symbol/boundary	10%
Attribute completeness	Many NULL/unknown	Minor gaps	Complete required fields	10%
Cross-map consistency	Conflict	Not checked/neutral	Supports other evidence	5%
Validation status	Historical only	Field checked/sample pending	Sampled/lab confirmed	5%

Total score	Confidence class	Use status
>=80	High	Target ranking-д ашиглаж болно; field validation шаардлагатай хэвээр
60-79	Medium	Overlay/interpretation-д болгоомжтой ашиглана
40-59	Low	Context evidence; decision-grade ашиглахгүй
<40 or critical fail	Needs verification	Ашиглахаас өмнө дахин шалгах, field/lab/report validation шаардлагатай

19. Data gap register

Gap type	Impact	Required action	Owner / status field
Missing legend	Confidence бууруулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Unclear legend	Confidence бууруулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Poor scan quality	Confidence бууруулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Weak coordinate grid	Confidence бууруулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status

Insufficient GCP	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
High residual error	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
CRS uncertainty	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Symbol unreadable	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Attribute uncertain	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Feature duplicated across maps	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Conflicting feature location	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Scale too regional for local use	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Field validation required	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Lab confirmation required	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status
Report text needed to interpret map	Confidence буурууулна / handover-д анхааруулга болно	Review, re-georeference, validate with field/lab/report, or use as context only	gap_owner, due_date, status

Data gap register column	**Description**
gap_id	Unique ID, e.g. BUD-GAP-001
source_map_id	Related BK-SCAN ID
feature_id	Related feature if spatial
gap_type	Controlled gap type
gap_description	Detailed issue
impact_on_use	How it affects use
confidence_impact	Decrease / hold / exclude
required_action	What to do
responsible_person	Owner
due_date	Target date
status	Open / In progress / Closed / Deferred
closure_note	How it was resolved

20. Cross-map integration

Vector evidence-үүдийг нэг map-аас авсан дан мэдээлэл гэж үзэхгүй. Дараах overlay / consistency check-ийг хийж confidence-д нөлөөлүүлнэ.

Overlay pair	Шалгах зүйл	Confidence impact	QGIS tool / method
2013 mineral occurrence points vs 2013 geology	Occurrence нь favorable lithology/contact/structure дээр байгаа эсэх	Match бол confidence өснө; mismatch бол data gap	Join attributes by location, Select by location
2013 prospectivity zones vs mineral occurrence points	Target zone дотор occurrence cluster байгаа эсэх	Давхцалтай бол P1/P2 ranking support	Count points in polygon
2013 source materials routes/observations vs occurrence points	Historic observation/sample/trench point давхцах эсэх	Field validation planning-д шууд ашиглана	Distance to nearest hub / buffer
1987 stream sediment anomalies vs geology	Anomaly drainage favorable lithology/structure-аас эхтэй эсэх	Drainage source confidence нэмэгдэнэ	Overlay + DEM drainage analysis
1987 heavy mineral anomalies vs DEM/drainage/geology	Indicator mineral source direction logical эсэх	Orientation soil/stream sediment planning support	Drainage catchment overlay
1987 mineral resources vs 2013 mineral occurrence	Historical occurrence continuity байгаа эсэх	Cross-source support нэмэгдэнэ	Spatial join + attribute compare
2013 metallogenic scheme vs 2020/2021 regional metallogenic map	Ore formation / belt consistency	Deposit model support	Overlay and narrative register
All evidence vs license boundary and buffers	License дотор/гадна/буферийн байрлал	Field work priority болон use limit тогтооно	Clip, intersection, distance

Cross-map consistency field	Description
evidence_id	Feature/evidence ID
related_map_1	First map ID
related_map_2	Second map ID
match_type	Overlap / Near / Conflict / Supports / No relation
spatial_relationship	Within / intersects / adjacent / upstream / downstream
geological_relationship	Contact / same unit / structure controlled / no relation
confidence_impact	Increase / Neutral / Decrease / Needs verification
required_action	Field check / review legend / re-georef / report check

21. Handover package ба acceptance criteria

Deliverable	File / folder	Use	Acceptance criteria
Master QGIS Project	XV023222_Buduunkhad_HistoricalScannedMaps_Vectorization_QGIS_EPSG32647_v02.qgz	Layer overlay, review, map production, QField package preparation	Opens without missing layers; relative paths OK
Master GeoPackage	XV023222_Buduunkhad_HistoricalScannedMaps_Vectorized_MasterGIS_EPSG32647_v02.gpkg	All vector evidence layers in EPSG:32647	All mandatory layers created or documented as N/A
Georeferenced GeoTIFF folder	04_Georeference_Check/02_Georeferenced_Rasters	Base raster for audit and future redigitizing	Priority map GeoTIFF complete + QAQC
Excel QA/QC workbook	XV023222_Buduunkhad_HistoricalScannedMaps_Vectorization_Register_QAQC_v02.xlsx	Register, QA/QC, confidence, data gap, change log	Required sheets complete; no missing required fields
PDF index maps	Phase1_Master_GIS_Index_Maps.pdf	Review / reporting	License + evidence layers shown
README + Change log	README.txt, Change_Log.xlsx	Audit trail and handover note	Version, CRS, source note, limitations documented

Handover phase	Required layers	Decision use
Phase 3 Geological / Metallogenic Synthesis	Geology, structures, metallogenic zones, occurrence context	Concept model, deposit style, regional context
Phase 4 Preliminary Prospect Ranking	Occurrence, target zones, geology, geochem anomalies, confidence layers	Evidence scoring, A/B/C/D ranking
Phase 6 Recon Mapping and pXRF	QField-ready observation/source materials/target points	Field validation and pXRF route planning
Phase 7 Rock Chip Sampling	Mineral occurrences, structures, target zones, source observations	Rock chip candidate locations

Deliverable	File / folder	Use	Acceptance criteria
Phase 8/9 Soil / Stream Sediment Planning	Drainage, heavy mineral, stream sediment anomaly, geology/structure	Orientation survey and systematic grid planning	
QField field validation package	Approved subset of points/targets/routes + forms	Mobile field verification	

22. Final workflow diagram

Raw Scanned Maps + Legends
  -> Working Copy + File Inventory
  -> Map-Legend Linkage Register
  -> CRS / Georeference Status Audit
  -> QGIS Georeferencing + GCP Table
  -> Georeferenced GeoTIFF EPSG:32647
  -> Georeference QA/QC + Raster Confidence Ranking
  -> Vector Digitizing by Map Type
  -> GeoPackage Layers + Source Traceability
  -> Attribute Domain Validation
  -> Topology / Geometry / Attribute QA/QC
  -> Excel Registers + QA/QC Workbook
  -> Cross-Map Evidence Integration
  -> Confidence Ranking + Data Gap Register
  -> Master GIS / QField / Prospect Ranking Handover

23. Appendices

Appendix A - Feature ID naming standard

Layer	ID prefix	Example
geology_units_50k_polygons	BUD-GEO50	BUD-GEO50-0001
geology_units_200k_polygons	BUD-GEO200	BUD-GEO200-0001
structures_faults_lines	BUD-STR	BUD-STR-0001
mineral_occurrences_points	BUD-MIN	BUD-MIN-0001
heavy_mineral_anomaly_polygons	BUD-HM-AN	BUD-HM-AN-0001
stream_sediment_anomaly_polygons	BUD-SS-AN	BUD-SS-AN-0001
prospectivity_target_zones_polygons	BUD-TGT	BUD-TGT-0001
source_material_observation_points	BUD-OBS	BUD-OBS-0001
source_material_route_lines	BUD-RTE	BUD-RTE-0001
metallogenic_zones_polygons	BUD-MET	BUD-MET-0001
data_gap_register_spatial	BUD-GAP	BUD-GAP-0001

Appendix B - QGIS Field Calculator expressions

Purpose	Expression	Тайлбар
UTM Easting	x($geometry)	Point layer-д x_utm

Purpose	Expression	Тайлбар
UTM Northing	y($geometry)	Point layer-д y_utm
Longitude WGS84	x(transform($geometry, @layer_crs, 'EPSG:4326'))	Point geometry longitude
Latitude WGS84	y(transform($geometry, @layer_crs, 'EPSG:4326'))	Point geometry latitude
Polygon area km2	area($geometry) / 1000000	Target/geology polygon талбай
Line length km	length($geometry) / 1000	Route/fault/structure length
Feature ID example	concat('BUD-MIN-', lpad(@row_number,4,'0'))	Draft ID үүсгэх

Appendix C - QField package preparation note

- QField-д зөвхөн Approved эсвэл Checked status-тэй local target-scale layers оруулна.
- Context-only 1:400k/1:500k metallogenic layer-ийг field package-д заавал оруулах шаардлагагүй; шаардлагатай бол non-editable reference layer болгоно.
- Field editable layer-үүд: source_material_observation_points, mineral_occurrences_points_validation_copy, prospectivity_target_zones_polygons_reference, source_material_route_lines_reference.
- Form tab: Source / Geometry / Observation / Mineralization / Confidence / Photo / QAQC.
- fid, system fields, geometry fields-ийг user form дээр hide хийнэ. Required fields-д constraint тавина.

Appendix D - Чанарын хяналтын анхааруулга

- Historical scanned map-derived vector data нь хүдэржилтийг батлах decision-grade evidence биш.
- Field validation, rock chip sampling, pXRF screening, laboratory assay шаардлагатай.
- Georeferenced scan map-ийн positional accuracy-г confidence flag-гүй ашиглаж болохгүй.
- Regional scale map-аас гарсан polygon/line-г local target boundary мэт ашиглаж болохгүй.
- Raster дээрх map symbol-ийг vector болгохдоо source_file, source_map_id, original_symbol, original_label талбаруудыг заавал бөглөнө.
- Raw scan file-г overwrite хийхгүй. Version update бүрт v02, v03 гэх мэтээр нэмэгдүүлнэ.

Critical QA/QC Notes

- Raw data-г засварлахгүй; processing copy дээр ажиллана.
- Final deliverables-ийн CRS нь EPSG:32647; native/raw CRS-г metadata-д хадгална.
- .tfw, .aux.xml, .ovr, .rpc, .eph, .txt зэрэг sidecar файлуудыг parent raster/image-тэй хамт хадгална.
- Scan map/georeferenced map-ийн positional accuracy-г residual болон confidence flag-тай хэрэглэнэ.
- ASTER final binary mask, Sentinel alteration ratio, KOMPSAT visual lineament, drone interpretation нь хүдэржилтийн баталгаа биш.
- pXRF нь lab assay-г орлохгүй; Au-ийн pXRF response-ийг decision-grade гэж үзэхгүй.
- CMCS/MRPAM nearest deposit нь contextual evidence бөгөөд тухайн license дотор хүдэржилт байгаа эсэхийг шууд батлахгүй.
- Final target confidence нь хээрийн шалгалт, дээжлэлт, лабораторийн assay, structural/geological evidence, шаардлагатай бол trench/geophysics/scout drilling-аар баталгаажна.

Methodology Limitation

- Энэ баримт бичиг нь methodology / workflow guide бөгөөд бодит raster processing, vector digitizing, assay interpretation, target grade estimation хийгээгүй.
- 78 raw input file-ийн агуулга, spatial correctness, coordinate accuracy, file integrity нь Phase 1 Data Audit-ээр баталгаажсаны дараа л analysis-ready гэж үзнэ.
- Remote sensing index болон alteration proxy нь surface condition, vegetation, shadow, weathering, sensor limitations, CRS/pixel alignment-аас хамаарч буруу response өгөх боломжтой.
- Historical map scale (1:500,000, 1:200,000, 1:50,000) нь target-scale field decision-д шууд хангалтгүй; field verification шаардлагатай.
- Эцсийн follow-up болон scout drilling decision нь энэ document-ийн workflow-оос гадна permit, land access, HSE, budget, environmental and legal requirements-тай нийцэх ёстой.